Online Supplement to

Fair vehicle routing with random route durations: ex-post ordered fairness, rank prices and risk-consistent tours

Quang-Vinh Dang, Hoang-Viet Vu, Phuc-Son Nguyen

This supplement contains the detailed pseudocode and parameters of the algorithm (Section S1), the complete computational study with all seventeen experiments E1–E17 and their tables and figures (Section S2), and per-instance results (Section S3). Numbers of sections, propositions, theorems, tables and figures without the prefix S refer to the main paper.

S1 Algorithm details

Algorithm S1 details the improvement step of the main loop (Algorithm 1 of the main paper), and Table S1 lists all parameters; they are identical for all methods and instances.

S2 Extended computational study

S2.1 Set-up

Instances. We follow the slicing rule of the deterministic fair routing literature (Ljubić, Puerto, & Torrejón, 2026). Block b of size n consists of the depot and the n consecutive customers $bn+1,\dots,(b+1)n$ of CVRPLIB X-n641-k35 (Uchoa et al., 2017), with ceiled Euclidean distances, $m=5$ vehicles and capacity $Q=\lfloor\sum_i q_i/4\rfloor$, so one vehicle's worth of slack is available. The main grid uses $n\in\{15,20,25,30,35\}$ and blocks $b=0,\dots,4$ (25 instances); the scale test uses $n\in\{50,75,100\}$ and blocks 0–5; the exactness test uses ten instances with $n=10$, $m=3$ and $Q=\lfloor\sum_i q_i/2.4\rfloor$.

Uncertainty. Unless stated otherwise $v=0.3$ (gamma arc noise) and $s_Y=0.15$. In the congestion experiment (E8) the arc coefficient of variation is vH_{ij}, where $H_{ij}\in[1/3,2]$ is largest for arcs near a randomly drawn congestion centre, so noise is no longer proportional to length.

Budget. C^* is the cost of the best of three restarts of the cost-minimising search; the total length of every fair plan must satisfy $\sum_k L_k \le B\,C^*$ with $B\in\{1.05,1.10\}$.

Methods. All methods share the search of Algorithm 1, the start plan, the time limit (15 s for $n\le 25$, 25 s for $n\ge 30$, 120 s for $n\ge 50$) and the random seed. They differ only in the objective and in the lower level. *Det* minimises the deterministic ordered measure $\ell_\lambda(L)$ of the tour lengths, i.e. the model of Ljubić et al. (2026) solved heuristically. *SAA* minimises the sample average of $\ell_\lambda(X)$ over 200 fixed arc scenarios. *SORTIE-0* minimises the exact ex-post value by PBOI with TSP-optimal tours ($\kappa=0$); *SORTIE-1* uses the risk-consistent lower level with $\kappa=1$. In Det and SAA the moves are screened by the deterministic and the sample objective respectively.

Algorithm S1 IMPROVESTEP(P, ϖ, ψ): screen, evaluate, confirm

for every relocate $(i : a \to b)$ and swap $(i \in a,\ j \in b)$ **do**
  compute insertion and removal deltas $\Delta L_a, \Delta L_b, \Delta W_a, \Delta W_b$
  if capacity holds and $\sum_k L_k + \Delta L_a + \Delta L_b \le (1 + \epsilon_B)\, B\, C^*$ **then**
    $\hat{\Delta} \leftarrow \varpi_a \Delta L_a + \varpi_b \Delta L_b + \psi_a \Delta\sigma_a + \psi_b \Delta\sigma_b$ ▷ eq. the prediction formula of Section 5.3
  end if
end for
$\mathcal{K} \leftarrow$ the K moves with the smallest $\hat{\Delta}$
for each move in $\mathcal{K}$ **do**
  re-solve routes a, b with *screening* tours; evaluate F by PBOI
end for
for each improving candidate, best first **do**
  re-solve routes a, b with the *exact* lower level; re-evaluate F
  if F still improves and $\sum_k L_k \le B\, C^*$ **then return** the confirmed plan
  end if
end for
return NONE

Table S1: Parameters of SORTIE (identical for all methods and instances).

Component	Parameter	Value
Lower level	exact Held–Karp up to	15 customers
	local-search starts beyond that	8
	screening tours	1 descent, ≤24 tours if $\kappa > 0$
Evaluator	Gauss–Legendre points per panel / quantile levels	6 / 13
	dispersion-price step	$0.05\,\sigma_k$
Screen	candidates evaluated per step K	8
	budget slack in the screen ϵ_B	2%
Ruin	removed customers q	2 to $\max(3, \min(12, \lceil n/4 \rceil))$
	price-weighted pool / related removal	probability 1/2 each
Recreate	cost tie-breaker in $s_b(i)$	10^{-3}
Repair	maximum cost-descent steps	200
Annealing	initial temperature / final ratio	$0.02\,\lvert F(P_0)\rvert$ / 10^{-3}

Evaluation. Every plan is evaluated out of sample by arc-level Monte Carlo with 20 000 scenarios (10 000 for $n \ge 50$), including the day factor, with common random numbers across methods. The reported *ex-post value* is the Monte Carlo estimate of $\mathbb{E}\ell_\lambda(\Phi)$. Paired comparisons use the Wilcoxon signed-rank test. All tables are generated from the raw results by the released scripts.

S2.2 Accuracy and cost of the evaluator (E1, E2)

Table S2 compares PBOI with 50 000-sample arc-level Monte Carlo on 168 plans: the cost-efficient plans and random feasible plans whose routes are sequenced by the lower level, for $n \in \{15, 25, 35, 50\}$. The mean absolute relative error is below 0.11% for all five measures and at most 0.44% in the worst case, and 99–100% of the differences lie within three Monte Carlo standard errors. The mean z-scores are not exactly zero (up to 0.34 for the MAD): PBOI carries a systematic bias of a few hundredths of a percent, which we attribute to the moment-matched gamma marginal of each route (a sum of arc gammas with different scales is not exactly gamma). PBOI takes 0.44 ms per plan for $m = 5$, against 0.28 s for 20 000-scenario Monte Carlo. A 200-scenario SAA is about seven times faster than PBOI but is a noisy estimate. The advantage of PBOI is therefore that it is exact and noise-free at a cost compatible with thousands of

Table S2: E1: PBOI against arc-level Monte Carlo (50 000 samples) on 168 plans; z: difference over its Monte Carlo standard error.

Measure	Plans	Mean |err| (%)	Max |err| (%)	Share $\|z\| < 3$
Max	168	0.041	0.137	0.994
Range	168	0.062	0.249	1.000
GMD	168	0.057	0.241	1.000
MAD	168	0.064	0.272	0.988
Var	168	0.108	0.440	0.994

Table S3: E3: what each method believes its unfairness is, relative to the true ex-post value (Monte Carlo without the day factor, which is what each model represents), mean signed error over 50 runs per cell. Last column: largest absolute error of the PBOI prediction for any method's plan.

	Believed vs true (mean signed %)				PBOI
Measure	Det	SAA	SORTIE-0	SORTIE-1	max |err| %
Max	-7.97	-0.71	-0.01	-0.01	0.20
Range	-17.56	-0.99	+0.03	+0.02	0.70
GMD	-14.68	-0.91	+0.04	+0.05	0.61
MAD	-16.17	-0.88	+0.05	+0.07	0.68
Var	-27.23	-1.33	+0.00	+0.02	0.98

evaluations per second, not raw speed.

On the ten $n = 10$ instances we enumerated all capacity- and budget-feasible partitions (4 to 145 per instance) with exact Held–Karp tours. The cost search found the optimal C^* on all ten. For both range and GMD, SORTIE-0 found the plan with the smallest exact ex-post value on nine of ten instances (mean gap 0.39%, maximum 3.9%). Det found the deterministic optimum on nine of ten; on the tenth instance, which has only 20 budget-feasible partitions, it stopped at a plan with twice the optimal deterministic range.

S2.3 The fairness illusion and the value of ex-post optimisation (E3)

Table S3 quantifies Proposition 9. A deterministic fair plan believes its unfairness is the ex-ante value, which understates the true ex-post value by 8% for the maximum, 15–18% for the range, GMD and MAD, and 27% for the variance. The in-sample SAA value is optimistic by about 1%, which is the optimiser's curse on 200 scenarios. The PBOI value believed by SORTIE has a mean signed error below 0.1%, a mean absolute error of 0.07–0.23% and a worst case of 1%.

Table S4 reports the out-of-sample ex-post value relative to Det. The signed-rank test favours SORTIE-0 and SORTIE-1 over Det in 9 of the 10 cells each. Pooled over all 250 runs, SORTIE-0 improves on Det by 0.89% on average (median 0.23%; better in 141 runs, tied in 96, worse in 13; $p < 10^{-18}$) and SORTIE-1 by 0.77% (better in 154, worse in 31). SAA improves by only 0.14% (median zero; better in 115, worse in 74) and is favoured significantly in 3 cells. Head to head, SORTIE-0 beats SAA in 145 runs and loses in 15, and SORTIE-1 beats it in 147 and loses in 27 ($p < 10^{-17}$). The largest gains are for the maximum at $B = 1.10$ (2.1%, 24 of 25 instances), where the extra budget lets the planner trade a slightly longer but less variable longest route.

The improvements are small, about one percent on average, and this is itself a finding. With homogeneous noise ($\sigma_k^2 = v^2 W_k$) the standard deviation of a route grows with its length, so the plans that balance lengths also nearly balance risks, and a deterministic fair plan is close to ex-post optimal even though its *assessment* of its own fairness is badly wrong. For context, the cost-optimal plan has an ex-post range of 2087 against about 1340 for the fair plans, at a cost 6.5% higher: fairness itself is worth optimising; the stochastic refinement adds about one

Table S4: E3: improvement of the out-of-sample ex-post value over Det (mean %, positive is better; in brackets the number of instances on which the method is strictly better). *: Wilcoxon signed-rank $p < 0.05$ in favour of the method; the test is on ranks and can disagree in sign with the mean when one instance dominates (Var, $B = 1.10$, SORTIE-1).

Measure	B	SAA	SORTIE-0	SORTIE-1
GMD	1.05	-0.40 (6/25)	-0.09 (8/25)	-0.03 (9/25)
GMD	1.10	-0.32 (9/25)	+0.94* (14/25)	+0.34* (17/25)
MAD	1.05	+0.86* (11/25)	+1.10* (13/25)	+1.24* (12/25)
MAD	1.10	+0.70 (13/25)	+0.96* (14/25)	+1.53* (18/25)
Max	1.05	+0.71* (14/25)	+0.61* (15/25)	+0.77* (15/25)
Max	1.10	+1.70* (23/25)	+2.08* (24/25)	+1.93* (24/25)
Range	1.05	+0.12 (13/25)	+0.64* (15/25)	+1.56* (15/25)
Range	1.10	-0.21 (13/25)	+1.07* (17/25)	+0.39* (15/25)
Var	1.05	+0.48 (7/25)	+0.50* (9/25)	+0.48* (13/25)
Var	1.10	-2.25 (6/25)	+1.09* (12/25)	-0.49* (16/25)

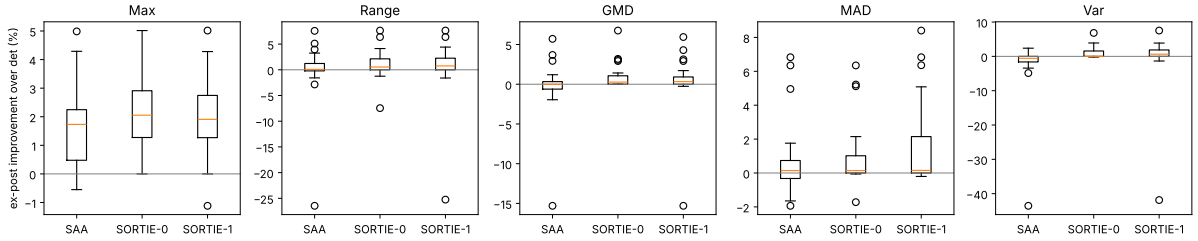


Figure S1: E3, $B = 1.10$: per-instance improvement of the ex-post value over Det.

percent on top. For range-optimised plans the $\mathrm{CVaR}_{0.9}$ of the realised range, which none of the methods optimises, is 2054 for Det, 2059 for SAA, 2037 for SORTIE-0 and 2032 for SORTIE-1; both SORTIE variants improve it at both budgets, SAA at neither.

An illustrative instance. Figure S2 shows the cost-optimal plan and the range-optimised Det and SORTIE-0 plans of X641-n30-b0 at $B = 1.10$, an instance whose SORTIE-0 gain (+0.47%) is close to the median of the 25 instances (+0.54%). The cost plan leaves one driver with a route of length 148 and another with 2610. Both fair plans balance the lengths to a deterministic range of about 940, and they differ only in how two route pairs are split. Figure S3 shows what the drivers experience. The realised workloads are wide and right-skewed, and the expected realised range is about 1225 for both fair plans: 30% above the deterministic range the Det planner reports. Figure S4 gives the distribution of the realised range of the two plans; in one day out of ten it exceeds 1750. The worst instance for SORTIE-0 in this cell is X641-n35-b0, where Det is 7.5% better (Figure S21 in Appendix S3); this is consistent with the search noise documented in E7b rather than a systematic effect, since on the same instance SORTIE-0 is better or tied in eight of the other nine cells.

How tight is the illusion bound? Figure S5 plots the gap between the true ex-post value and the ex-ante value of every Det plan against the bound of Proposition 9. No plan violates the bound. The bound is conservative: the realised gap is on average 33% of the bound for the maximum, 18% for the range and 11–12% for the GMD and MAD, because it charges every route its full standard deviation while only the routes near the extreme ranks matter. Its value is that it is computable from the quadratic lengths alone and grows like $v\sqrt{\sum_k W_k}$, which is the scaling observed in E6.

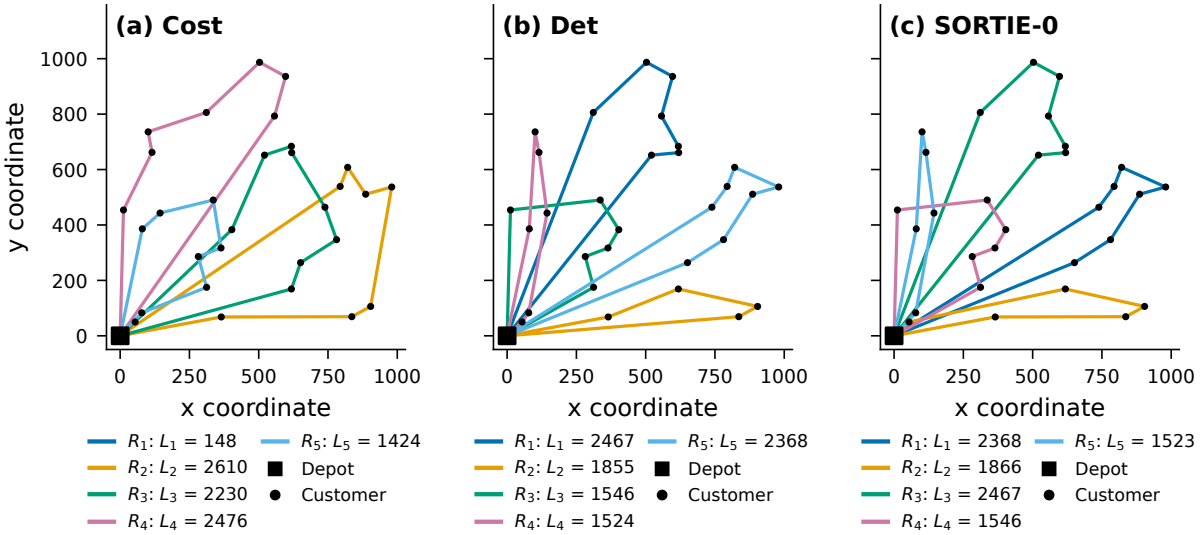


Figure S2: X641-n30-b0, $B = 1.10$: cost-optimal plan and the range-optimised Det and SORTIE-0 plans. L_k: tour lengths.

S2.4 Route consistency (E4)

E4 removes the lower level: the planner may sequence each route freely (*free*) instead of using the TSP-optimal tour (*lower*). For the maximum, free sequencing changes almost nothing, as monotonicity predicts in the deterministic case. For the range and the GMD the free planner lengthens the short routes: 23 of 25 Det plans and 24 of 25 SORTIE plans for the range contain a route that is not a shortest tour of its customers; the excess length is 95 (Det) and 113 (SORTIE) distance units on average over the runs that have one. This lowers the ex-post value by 1–3% for Det on both measures and for SORTIE on the range (Table S5); for SORTIE on the GMD it is 0.9% worse, because the larger space slows the search. Wherever it helps, the reduction comes from making some drivers drive detours. This is exactly the behaviour signalled by negative rank prices and excluded by the bilevel constraint. The deterministic belief error persists with free tours (−24% for the range); the PBOI belief remains exact.

S2.5 Rank-price screening (E5)

Replacing the price screen by $K = 8$ random feasible moves, with everything else unchanged, gives slightly more iterations (3200 against 2700) but worse plans: price screening lowers the ex-post value by 5.6% for the range and 8.5% for the GMD on average (medians 0% and 5.4%; better on 12 and 17 of 25 instances, worse on 3 and 2, ties otherwise; $p = 0.005$ and $p < 10^{-3}$). For the range the mean is driven by a few instances on which random screening stalls far from a good plan; for the GMD the advantage is broad. The rank prices are thus an effective first-order guide for the search, and not only an interpretive device.

S2.6 Noise level (E6)

Table S6 varies v for $n = 25$. The illusion grows monotonically with v, as the bound of Proposition 9 suggests: at $v = 0.5$ a deterministic plan understates its range by 43%. The gain of SORTIE over Det also grows with v for the range, from −0.5% at $v = 0.1$ (where the illusion is small and the differences are search noise) to 2.1% at $v = 0.5$ (better on all five instances); for the GMD it is 1.1%, 0.7% and 1.8%.

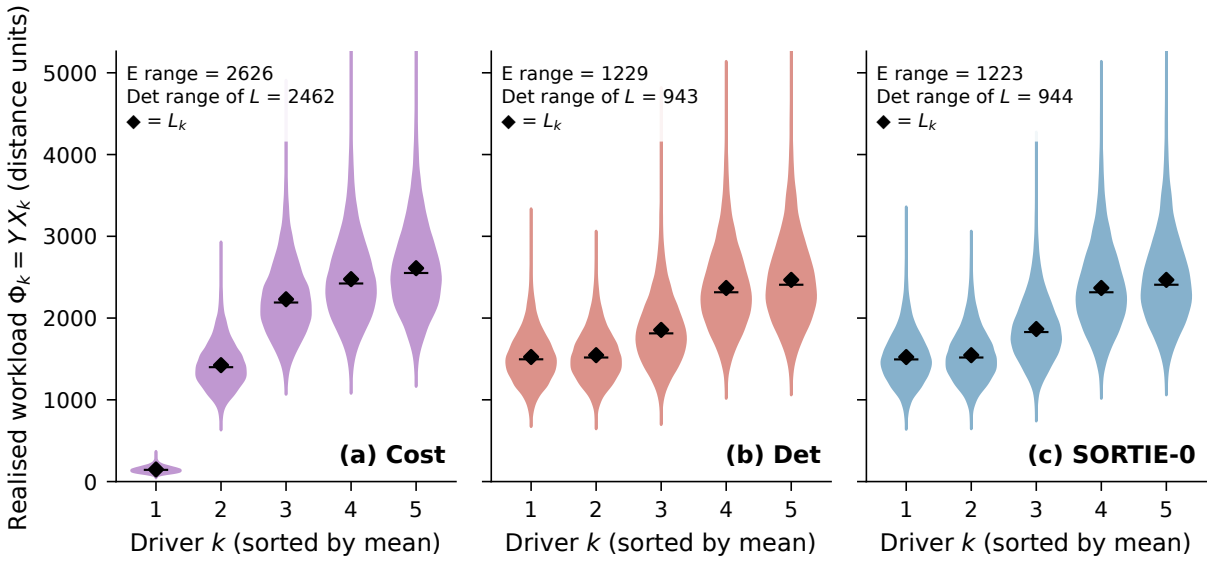


Figure S3: Simulated realised workloads $\Phi_k = YX_k$ of the three plans of Figure S2 (20 000 days); diamonds mark the tour lengths L_k that a deterministic model sees.

S2.7 Larger instances (E7, E7b)

E7 runs $n \in \{50, 75, 100\}$ on six blocks each, with a 120 s limit, for the range, GMD and maximum (18 instance–measure pairs per size). With the screening lower level the search runs about 9000 iterations per run at $n = 50$ and 3500 at $n = 100$ for SORTIE-0 (Det: 15 000 and 4400). PBOI remains accurate (mean absolute error 0.15%, at most 0.9%) and the deterministic belief error is -17% on average, as on the small instances.

Table S7 pools the 18 pairs per size. SORTIE-0 improves on Det by about 2% at every size and is better on 12–13 of 18 pairs; the improvement is significant at $n = 50$ ($p = 0.035$) but not at $n = 75$ or $n = 100$ ($p = 0.17$). SORTIE-1 is significant at $n = 50$ but not beyond: its lower level runs a dichotomic search per route and completes only about 850 iterations at $n = 100$, and 56% of its runs there (39% at $n = 75$) still improved by more than 0.5% in the second half of the time limit. Single runs at $n \geq 75$ are thus dominated by search noise, especially for the range and the GMD, whose value depends on a few routes.

E7b separates the method from the search noise at $n = 100$: the six instances and three measures are solved by Det, SAA and SORTIE-0 with three solver seeds each and a 240 s limit, and each method is represented by the mean ex-post value over its seeds (Table S8). Averaged over seeds, SORTIE-0 improves on Det by 4.2% (median 2.5%, better on 13 of 18 pairs, $p = 0.021$); SAA improves by 3.2% ($p = 0.10$). The difference between SORTIE-0 and SAA (1.1%, 12 of 18) is not significant ($p = 0.25$), and no single measure is significant on six instances. The seed-to-seed variation is itself informative: for the range, the ex-post value of the Det plan varies by 11% across seeds, that of SORTIE-0 by 3%. The deterministic range is flat over many plans whose realised fairness differs, and the search picks among them arbitrarily; the ex-post objective distinguishes them. Even at 240 s, 19% of the SORTIE-0 runs improved their in-model value by more than 0.5% in the second half of the run, so the absolute values at $n = 100$ are not fully converged.

Figure S8 shows the anytime behaviour behind these numbers, and Figure S9 the seed-to-seed spread at $n = 100$.

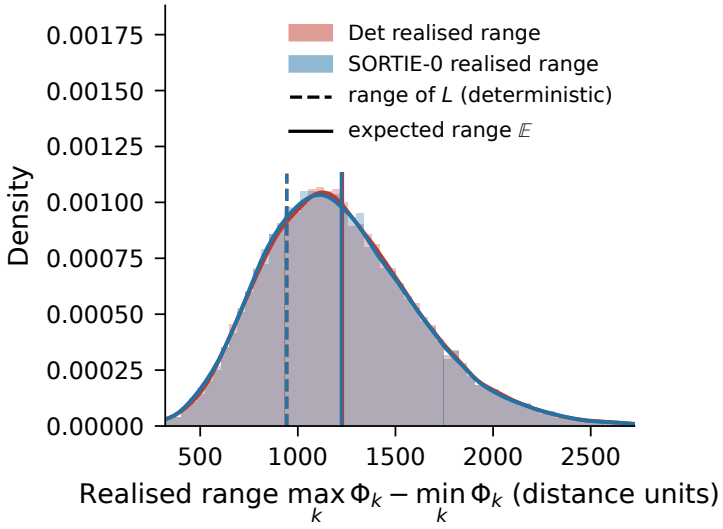


Figure S4: Distribution of the realised range of the Det and SORTIE-0 plans of Figure S2; dashed: deterministic range of the tour lengths, solid: expected realised range.

Table S5: E4: free tours against lower-level tours, $B = 1.10$, 25 instances per row. [a]Runs with at least one route that is not a shortest tour of its customers. [b]Mean number of such routes. Excess: total length above the shortest tours of the same customer sets. Belief err: believed minus true ex-post value. Ex-post: out-of-sample ex-post value.

Measure	Method	Runs[a]	Routes[b]	Excess	Belief err (%)	Ex-post
Max	Det-lower	0/25	0.00	0.0	-8.94	2221.0
Max	Det-free	0/25	0.00	0.0	-9.01	2232.4
Max	SORTIE-lower	0/25	0.00	0.0	-0.00	2174.8
Max	SORTIE-free	7/25	0.36	2.7	+0.00	2177.1
Range	Det-lower	0/25	0.00	0.0	-23.11	1099.7
Range	Det-free	23/25	1.00	87.8	-24.19	1088.5
Range	SORTIE-lower	0/25	0.00	0.0	+0.04	1098.4
Range	SORTIE-free	24/25	1.64	108.4	+0.06	1072.6
GMD	Det-lower	0/25	0.00	0.0	-19.46	437.7
GMD	Det-free	21/25	1.12	90.0	-20.02	425.9
GMD	SORTIE-lower	0/25	0.00	0.0	+0.07	434.5
GMD	SORTIE-free	23/25	1.52	82.4	+0.10	438.6

S2.8 Heterogeneous noise: congestion (E8)

E3 suggests that the stochastic refinement matters little when a route's variance is proportional to its length. E8 breaks that proportionality: arcs near a congestion centre have up to twice the base coefficient of variation and arcs far from it a third. We rerun the main grid ($B = 1.10$; range, GMD and max) on the congested versions of the same 25 instances, starting from the same cost plans.

The illusion becomes somewhat larger: a deterministic plan understates its range by 26%, its GMD by 22% and its maximum by 10%, against 23%, 19% and 9% on the same cells under homogeneous noise. The PBOI belief has a mean signed error below 0.2% (mean absolute error at most 0.26%). The gain of optimising ex-post fairness, however, grows only slightly (Table S9). Pooled over 75 runs, SORTIE-0 improves on Det by 1.51% (median 1.24%; better in 53, worse in 3; $p < 10^{-8}$) and SORTIE-1 by 1.66% (better in 57, worse in 10), against 1.36% for SORTIE-0 on the same cells without congestion. By measure the change is 0.94% → 1.40% for the GMD, 2.08% → 2.15% for the maximum and 1.07% → 0.96% for the range. SAA gains 1.28%, but

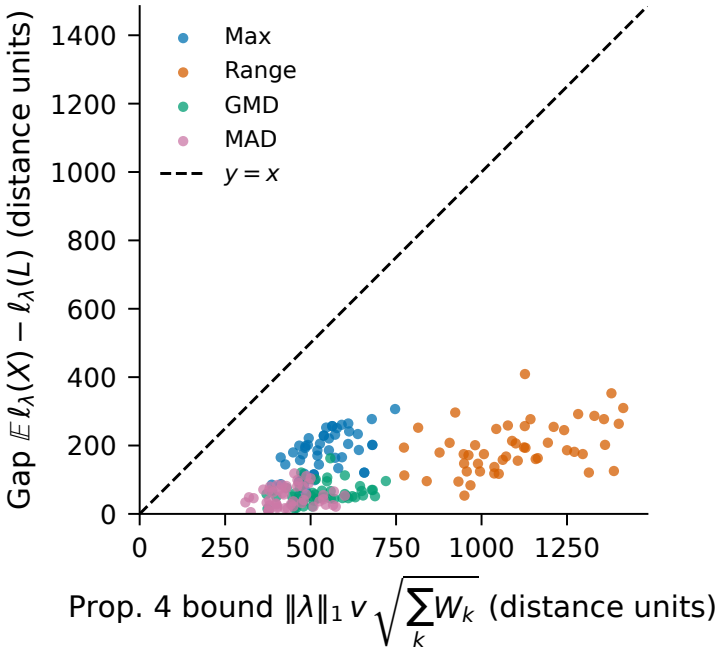


Figure S5: Fairness illusion of the Det plans of E3 (true ex-post minus ex-ante value) against the bound $\|\lambda\|_1 v(\sum_k W_k)^{1/2}$ of Proposition 9.

Table S6: E6: arc noise v, $n = 25$, $B = 1.10$, five instances per row.

v	Measure	Det belief err (%)	SORTIE-0 vs Det
0.1	GMD	-6.4	+1.05 (2/5)
0.1	Range	-8.7	-0.54 (1/5)
0.3	GMD	-22.0	+0.65 (2/5)
0.3	Range	-28.7	+0.96 (3/5)
0.5	GMD	-35.6	+1.84 (4/5)
0.5	Range	-42.7	+2.14 (5/5)

with a median of only 0.47%, and SORTIE-0 beats it head to head in 49 of 75 runs (worse in 6; $p < 10^{-6}$). The $\mathrm{CVaR}_{0.9}$ of the range is 1919 for Det, 1905 for SAA, 1912 for SORTIE-0 and 1907 for SORTIE-1. Heterogeneous noise thus makes a deterministic plan's self-assessment much worse, but in our instances it changes the best plan only a little more than homogeneous noise does: the congested arcs are concentrated in one part of the network, and the assignment that balances lengths still roughly balances risk there.

Figure S10 shows the congestion multiplier on the illustrative instance, and Figure S11 pairs every instance and measure with and without congestion: the points scatter around the diagonal, with the GMD above it and the range on both sides.

S2.9 The budget frontier (E9)

E9 repeats the comparison of Det and SORTIE-0 on the 10 instances with $n \in \{25, 35\}$ for budget factors B from 1.02 to 1.30 (Table S10, Figure S12). Two effects stand out. First, the gain of SORTIE-0 grows with the budget: from about 1% for $B \le 1.10$ to 3.2% at $B = 1.15$ and 6.8% at $B = 1.30$, where SORTIE-0 is better on all 30 instance–measure pairs. With little slack every fair plan is close to the cost-optimal one and there is little room to choose; with more slack there are many plans whose deterministic range is about equal but whose realised range differs, and only the ex-post objective can tell them apart. Second, for $B \ge 1.15$ SORTIE-0 uses *less* of the budget (below that the total lengths agree within 0.3 percentage points). At $B = 1.30$

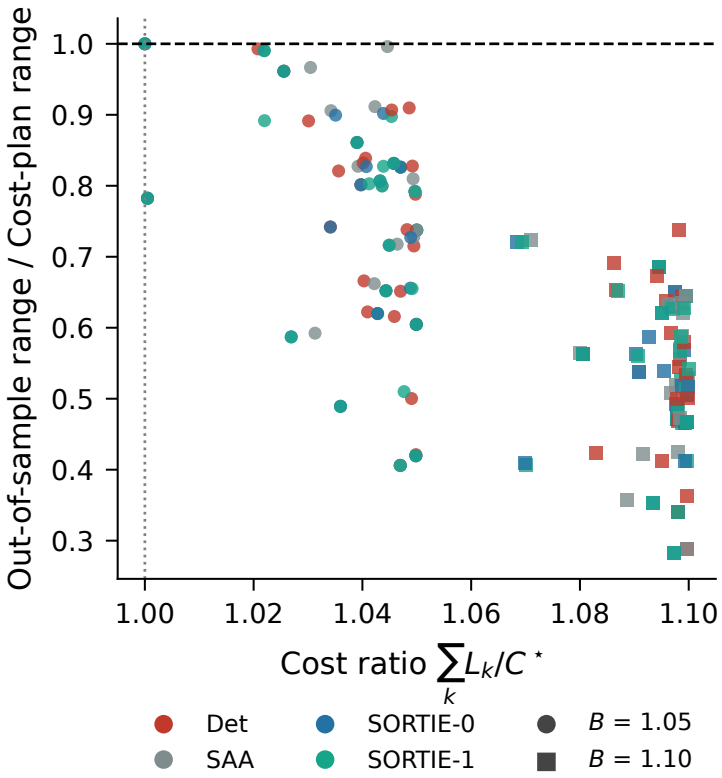


Figure S6: E3, range-optimised plans: total length relative to C^* against the expected realised range relative to that of the cost-optimal plan.

Table S7: E7: improvement of the ex-post value over Det, $B = 1.10$, 120 s, pooled over six instances and three measures (range, GMD, max); format as Table S4.

n	SAA	SORTIE-0	SORTIE-1
50	+1.56 (12/18)	+2.04* (13/18)	+1.20* (15/18)
75	+1.96 (11/18)	+2.04 (13/18)	-0.29 (11/18)
100	+1.52 (10/18)	+2.21 (12/18)	-0.04 (8/18)

the Det plans for the maximum have a total length 20.5% above C^* and an expected realised maximum 5.2% below that of the cost plan; the SORTIE-0 plans use 13.5% and achieve 8.7%. For the maximum the deterministic objective even deteriorates from $B = 1.20$ to $B = 1.30$: the extra length it buys goes into routes whose deterministic length is irrelevant to the maximum but whose variance raises the realised one. At every budget level the ex-post objective reaches lower realised unfairness, and for generous budgets it does so with less total length.

The same cells at $B = 1.10$ appear in E3, and the per-cell means differ (for example -0.7% for the range in E9 against $+0.9\%$ in E3 on the same ten instances). The runs are identical except for wall-clock timing: the search stops at a time limit, so the iteration at which it stops, and hence the plan, can differ between two executions on a shared machine. This is another manifestation of the search noise analysed in E7b, and the reason we draw conclusions only from pooled comparisons.

S2.10 Fleet size (E10)

E10 varies the number of vehicles on the $n = 35$ instances, with capacity $\lfloor \sum_i q_i/(m-1) \rfloor$ so that one vehicle's worth of slack remains (Table S11, Figure S13). The illusion grows with m: a deterministic plan understates its range by 10% with three vehicles, 24% with five and 27% with

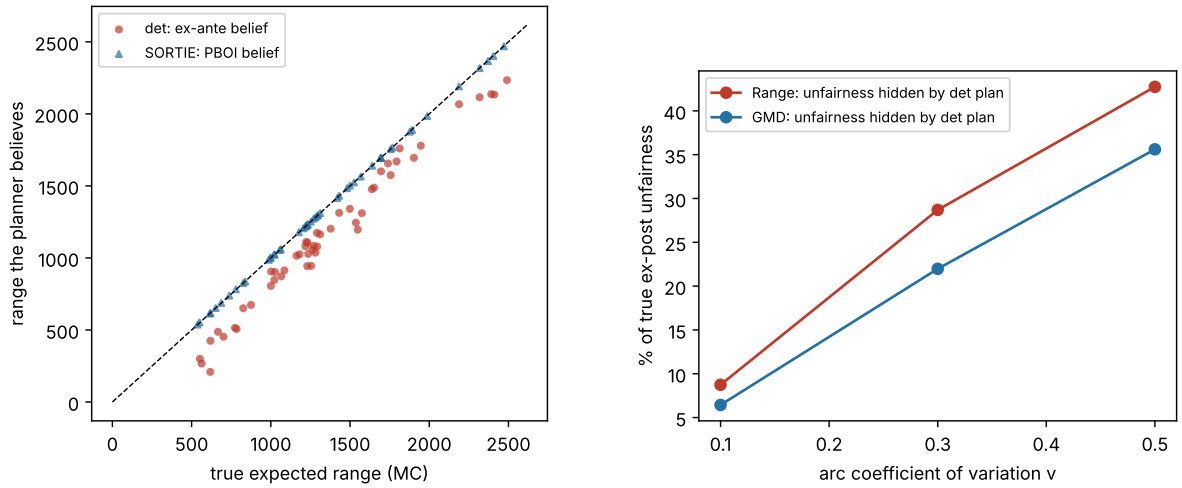


Figure S7: Left: range believed by Det (ex-ante) and by SORTIE (PBOI) against the true expected range, E3. Right: share of the true ex-post unfairness that a deterministic plan does not see, against v (E6).

Table S8: E7b, $n = 100$, $B = 1.10$, 240 s, three solver seeds: improvement of the seed-averaged ex-post value over Det (six instances per measure; format as Table S4) and coefficient of variation of the ex-post value across seeds, averaged over instances.

	vs Det, mean of 3 seeds		Seed CV (%)		
Measure	SAA	SORTIE-0	Det	SAA	SORTIE-0
Range	+6.45 (4/6)	+8.68 (5/6)	11.3	4.7	3.0
GMD	+2.49 (3/6)	+3.18 (4/6)	6.4	2.7	4.5
Max	+0.78 (4/6)	+0.86 (4/6)	1.4	0.8	1.4
All	+3.24 (11/18)	+4.24* (13/18)	SORTIE-0 vs SAA: +1.09 (12/18)		

seven. More routes compete for the extreme ranks, so more of the realised unfairness comes from which route happens to be longest on the day, which a deterministic model cannot see. The gain of SORTIE-0 is small with three and five vehicles (0.5% and 0.6%; with $m = 5$ not significant on these 15 pairs, $p = 0.06$) and 3.2% with seven (better on 13 of 15 pairs, $p = 0.001$). With $m = 3$ the GMD is proportional to the range (4/9 of it), so the two rows coincide.

S2.11 Common shocks in practice (E11)

E11 re-evaluates all 500 E3 plans with $B = 1.10$ under day-factor dispersions $s_Y \in \{0, 0.15, 0.3, 0.5\}$, using the same arc draws (Table S12, Figure S14). As Proposition 3 predicts, the expected ordered measures are unchanged: the ratio of the value with and without the day factor is 1.000 up to Monte Carlo error (at most 1.0007 on average, at $s_Y = 0.5$). The expected empirical variance is multiplied by 1.023, 1.095 and 1.285, against $e^{s_Y^2} = 1.023$, 1.094 and 1.284. The tail is a different matter: the $\mathrm{CVaR}_{0.9}$ of the realised range grows by 7%, 25% and 58%. A fleet-wide shock is irrelevant for the choice of a plan by expected ordered fairness, but not for the risk of a very unfair day.

S2.12 Robustness to the noise model (E12)

PBOI assumes independent route durations with given means and variances, and SORTIE was tuned to gamma arc noise. E12 re-evaluates the $B = 1.10$ plans of E3 (500 plans) under four arc-

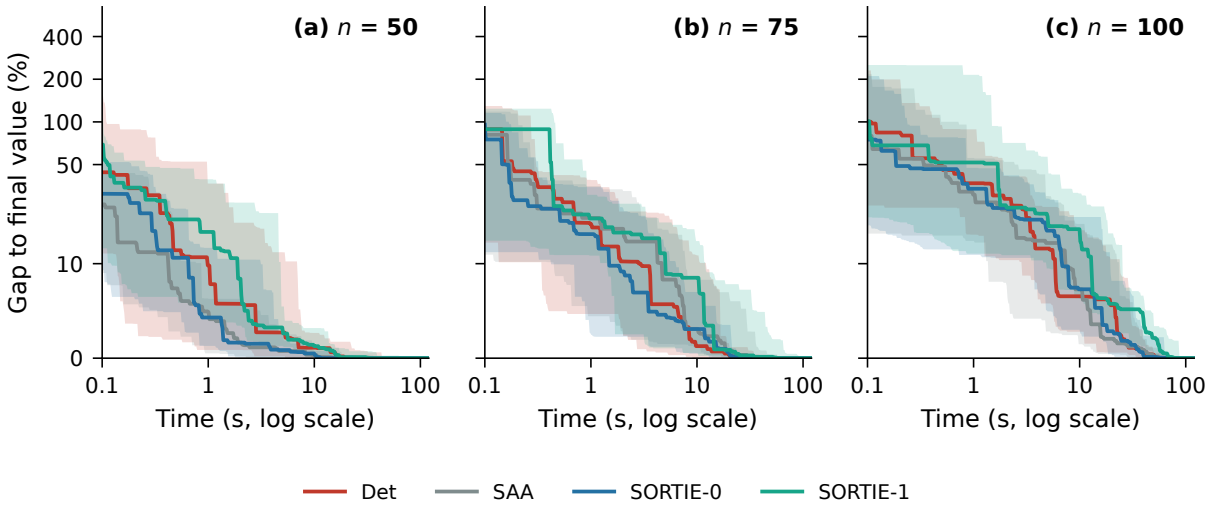


Figure S8: E7: best in-model value relative to the final value over the run (median and interquartile range over the 18 runs per method and size).

Table S9: E8: congested arc noise, $B = 1.10$, 25 instances per row; improvement of the ex-post value over Det (format as Table S4).

Measure	SAA	SORTIE-0	SORTIE-1
GMD	+0.76 (11/25)	+1.40* (13/25)	+1.82* (15/25)
Max	+1.87* (23/25)	+2.15* (23/25)	+2.21* (24/25)
Range	+1.22 (13/25)	+0.96* (17/25)	+0.94* (18/25)

noise families, and those of E8 (300 plans) under the first two (the other two cannot reach the low per-arc variability of the uncongested arcs of E8) with the same mean and coefficient of variation per arc: gamma (the planning model), lognormal (heavier right tail), *incident* (with probability 0.02 an arc takes 2.5 times as long, the rest of the variability being gamma), and *regional* (a multiplicative factor shared by all arcs whose tail lies in the same quadrant of the plane, which makes routes in the same region positively correlated and so violates the independence that PBOI assumes) (Table S13, Figure S15).

The accuracy of PBOI degrades gracefully. Under lognormal noise its mean absolute error rises from 0.16% to 0.29%; under incidents to 1.4%, with a bias of +1.4% (it overstates unfairness) and a worst case of 9.5%; under regional correlation to 1.9%, with a worst case of 20% for the variance measure, the one most affected by correlation. What matters for planning is that the ranking of plans survives: under every family SORTIE-0 remains significantly better than Det, by 1.0% to 1.2% on the E3 plans and 1.5% on the E8 plans (gamma and lognormal), against 1.2% and 1.5% under the planning model.

S2.13 Risk attitude of the drivers (E13)

E13 solves the $n \in \{25, 35\}$ instances with the risk-consistent lower level for $\kappa \in \{0, 0.5, 1, 2, 4\}$ (Table S14, Figure S16). Under homogeneous noise the risk-consistent tour rarely differs from the shortest one: even at $\kappa = 4$ only 14–16% of the routes change, and the extra length is about 13 distance units per plan, under 0.2% of the total. The ex-post value changes by less than the search noise (median ratio to $\kappa = 0$ equal to one, interquartile range within $\pm 1\%$). The reason is that, for a fixed customer set, a tour's length and its quadratic length are nearly comonotone: the shortest tour is also nearly the least variable one. Proposition 11 shows that this need not be so, and with heterogeneous noise the two can separate, but on these instances the choice of

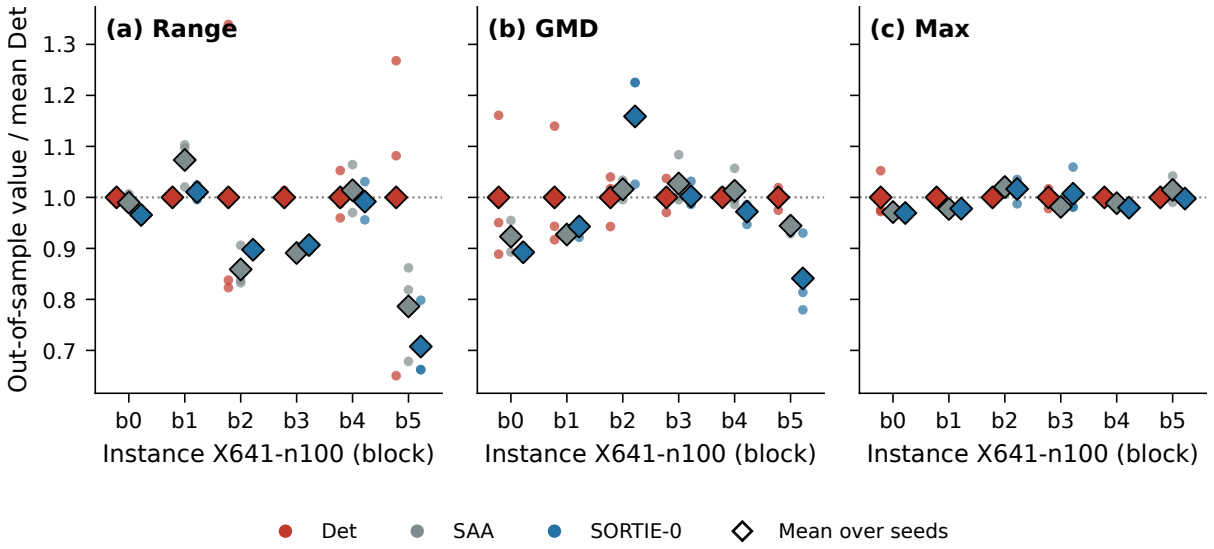


Figure S9: E7b, $n = 100$: ex-post value of each solver seed, relative to the mean Det value of the instance; larger markers: seed means.

Table S10: E9: improvement of the ex-post value of SORTIE-0 over Det by budget factor, $n \in \{25, 35\}$, ten instances per measure; cost ratio: mean $\sum_k L_k/C^*$ of the SORTIE-0 plans (format as Table S4).

B	Cost ratio	Pooled	Range	GMD	Max
1.02	1.014	+1.43* (17/30)	+1.40 (4/10)	+2.07 (5/10)	+0.82* (8/10)
1.05	1.041	+0.73* (19/30)	+0.81 (5/10)	+0.66 (5/10)	+0.73* (9/10)
1.10	1.094	+0.67* (20/30)	-0.72 (7/10)	+0.30 (3/10)	+2.45* (10/10)
1.15	1.134	+3.20* (27/30)	+5.92* (8/10)	+1.31 (9/10)	+2.37* (10/10)
1.20	1.170	+2.46* (26/30)	+2.59 (9/10)	+2.34* (7/10)	+2.46* (10/10)
1.30	1.222	+6.79* (30/30)	+7.64* (10/10)	+9.11* (10/10)	+3.63* (10/10)

κ is a matter of the drivers' preferences rather than of fairness.

S2.14 Why the rank prices work (E14)

E14 checks the first-order model behind the screen directly. For the cost-optimal start plan and the final SORTIE-0 plan of the ten $n \in \{25, 35\}$ instances, we evaluated up to 400 relocate and swap moves both by the rank-price prediction and exactly (exact lower level and PBOI); 5577 moves in total (Table S15, Figures S17 and S18). The prediction ranks the moves almost as the exact evaluation does: Spearman's ρ is 0.89–0.94 on the start plans and 0.78–0.89 on the final plans. On the start plans the exact-best improving move is among the 8 best predicted moves in 80% of the cases (random ordering: 7%) and always among the 21 best. On the final plans of SORTIE-0 there is no improving relocate or swap move in 28 of 30 cases: the search ends in genuine local optima of its neighbourhood. Consistently, the number of candidates K evaluated per step barely matters (Table S16): the median ex-post value is the same for K from 1 to 32.

S2.15 Is a larger sample enough? (E15)

A natural objection is that SAA with 200 scenarios is simply too small. E15 reruns SAA with $S \in \{50, 200, 1000, 5000\}$ scenarios on the $n \in \{25, 35\}$ instances (Table S17, Figure S19). SORTIE-0 is better than every SAA variant on average: by 1.4% ($S = 50$), 1.6% ($S = 200$), 1.1% ($S = 1000$) and 0.9% ($S = 5000$), significantly for $S \leq 1000$. The gap shrinks as the sample

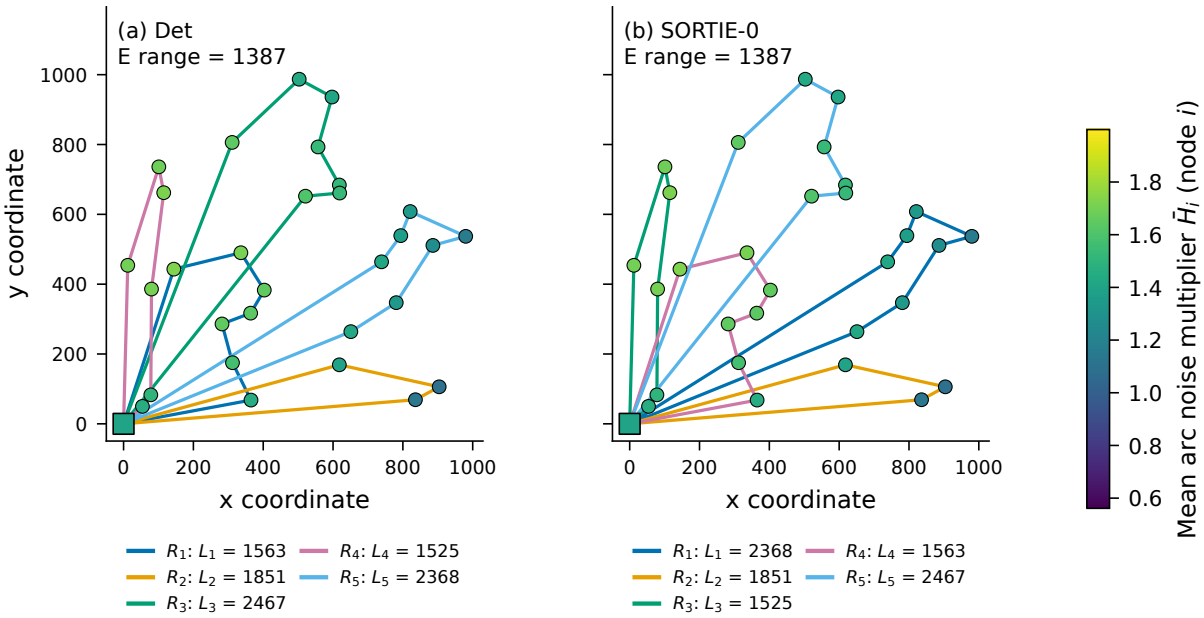


Figure S10: X641-n30-b0 with congestion: customers coloured by the mean noise multiplier H of their arcs, with the range-optimised Det and SORTIE-0 plans of E8.

Table S11: E10: fleet size, $n = 35$, five instances per measure, $B = 1.10$ (format as Table S4).

m	Measure	Det belief err. (%)	SAA vs Det	SORTIE-0 vs Det
3	Pooled	-8.2	+2.83* (7/15)	+0.48* (8/15)
3	Range	-9.9	+3.66 (2/5)	+0.14 (2/5)
3	GMD	-9.9	+3.66 (2/5)	+0.14 (2/5)
3	Max	-4.6	+1.17 (3/5)	+1.17 (4/5)
5	Pooled	-18.6	-1.62 (8/15)	+0.64 (9/15)
5	Range	-24.1	-3.11 (2/5)	+2.78 (3/5)
5	GMD	-23.2	-4.01 (1/5)	-3.58 (1/5)
5	Max	-8.5	+2.26 (5/5)	+2.71 (5/5)
7	Pooled	-20.3	+2.34* (10/15)	+3.16* (13/15)
7	Range	-26.7	+1.36 (3/5)	+4.58 (5/5)
7	GMD	-23.3	+3.34 (2/5)	+2.14 (3/5)
7	Max	-10.9	+2.31 (5/5)	+2.76 (5/5)

grows, as it should, since SAA converges to the objective that PBOI evaluates exactly; with 5000 scenarios SAA is still worse on 9 pairs and better on 2 (the rest tied), but the difference is no longer significant at the 5% level ($p = 0.0505$). The sample average is cached per route, so the number of iterations drops only from about 5300 to 4200, and the remaining gap is the sampling error of the objective, not a lack of search time.

S2.16 Simpler stochastic baselines (E16)

Two simpler ways of taking uncertainty into account are tested on the full E3 grid at $B = 1.10$ (Table S18). *Det-σ* applies the deterministic ordered measure to risk-adjusted lengths $L_k + \sigma_k$, the fairness analogue of the robust-length heuristics common in stochastic routing. It is *worse* than Det (-0.8%, better on 37 and worse on 61 of 125 pairs, $p = 0.02$), and SORTIE-0 beats it by 1.9% (97 of 125, $p < 10^{-14}$). Adding a standard deviation to every route treats each route's risk separately, whereas the expected ordered measure depends on how likely each route is to occupy each rank, which is what the Poisson–binomial integral and the rank prices capture. *PBOI-*

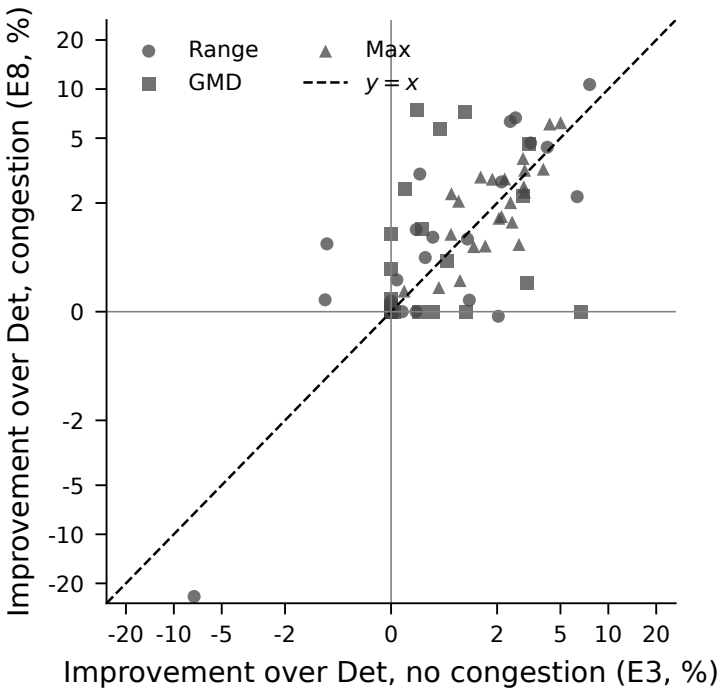


Figure S11: Improvement of SORTIE-0 over Det per instance and measure, without (E3) and with (E8) congestion, $B = 1.10$.

Table S12: E11: ratio of each measure with the day factor to its value without, mean ± standard deviation over the 500 E3 plans with $B = 1.10$.

s_Y	Max	Range	GMD	MAD	Var	CVaR90 range	$e^{s_Y^2}$
0.00	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000
0.15	1.000 ± 0.001	1.000 ± 0.001	1.000 ± 0.001	1.000 ± 0.001	1.023 ± 0.002	1.074 ± 0.016	1.023
0.30	1.000 ± 0.002	1.000 ± 0.002	1.000 ± 0.002	1.000 ± 0.002	1.095 ± 0.005	1.251 ± 0.042	1.094
0.50	1.001 ± 0.003	1.001 ± 0.003	1.001 ± 0.003	1.001 ± 0.003	1.285 ± 0.011	1.579 ± 0.081	1.284

N replaces the gamma marginals by normal ones; it performs like SORTIE-0 (no significant difference, $p = 0.65$), so the method is insensitive to the marginal family once the first two moments are right.

S2.17 A second benchmark family (E17)

All instances so far are slices of one file, and slicing places the depot at the first node of each block. E17 tests twelve further CVRPLIB X instances of Uchoa et al. (2017) (X-n101-k25 to X-n153-k22), which were generated with different depot positions and customer distributions. From each we take the depot of the file and its first 50 customers, keep $m = 5$ and the capacity rule $Q = \lfloor \sum_i q_i/4 \rfloor$, and solve range, GMD and maximum at $B = 1.10$ with Det, SAA, SORTIE-0 and Det-σ, 120 s and two solver seeds; each method is represented by the mean ex-post value over its seeds (Tables S19 and S20, Figure S20).

The results are stronger than on the sliced instances. The illusion reaches 40% for the range and 35% for the GMD, against 23% and 19% on the sliced instances at the same budget; plausibly the file's own depot, which is often eccentric, makes the routes differ more in variance than a block depot does. SORTIE-0 improves on Det by 2.2% on average (median 1.7%; better on 31 and worse on 5 of 36 pairs, $p < 10^{-6}$), with a positive average on every one of the twelve instances (0.7% to 4.2%). SAA improves by 0.7%, and SORTIE-0 beats it on 33 of 36 pairs ($p < 10^{-7}$). Det-σ is again worse than Det on average (−0.8%; better on 13 and worse on 23

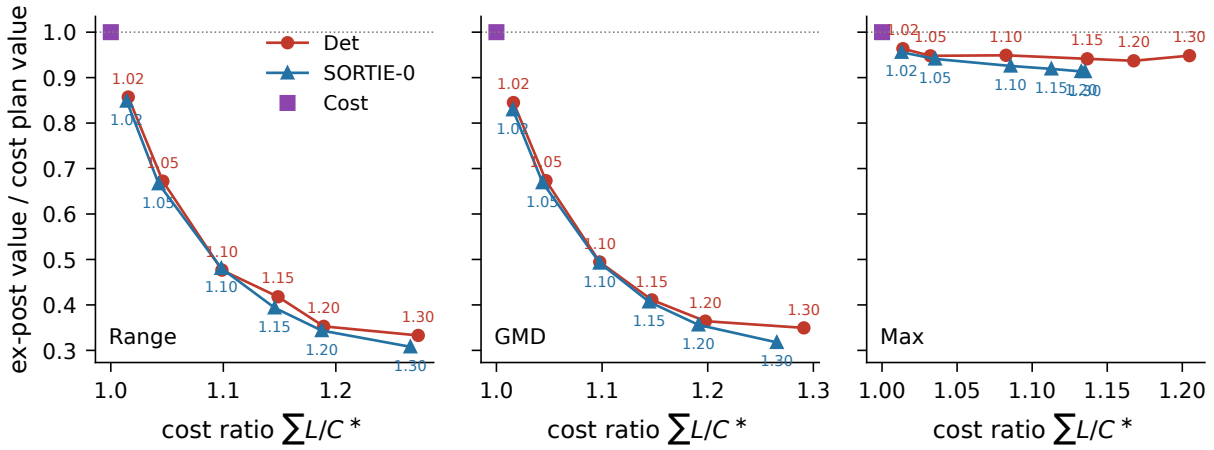


Figure S12: E9: price of fairness. Mean total length relative to C^* against the mean expected realised unfairness relative to the cost-optimal plan, for budget factors B from 1.02 to 1.30 (labels).

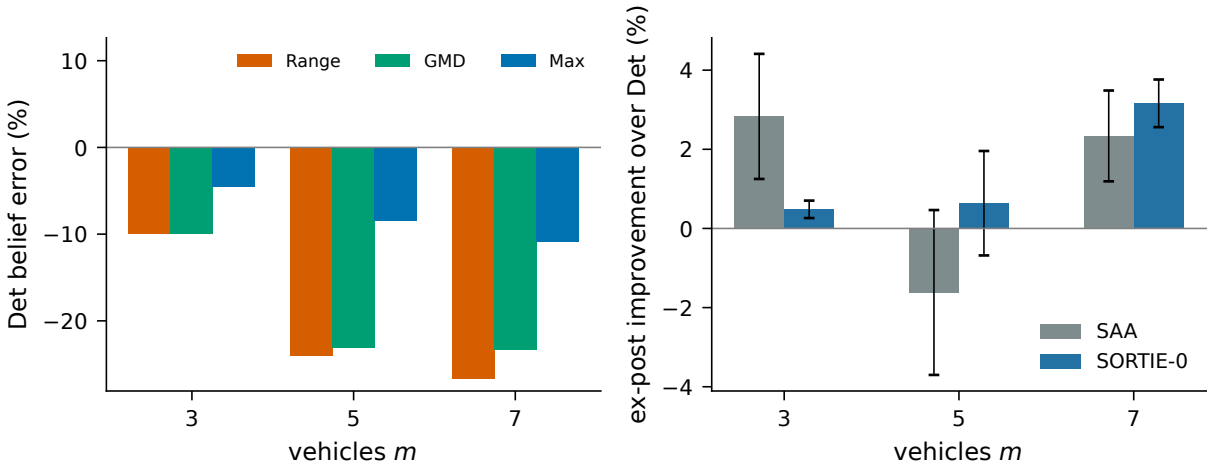


Figure S13: E10: belief error of Det and improvement over Det against the number of vehicles.

pairs, not significant here, $p = 0.15$). The conclusions of E3 and E16 thus carry over to instances with other depot positions and customer distributions, with larger effects.

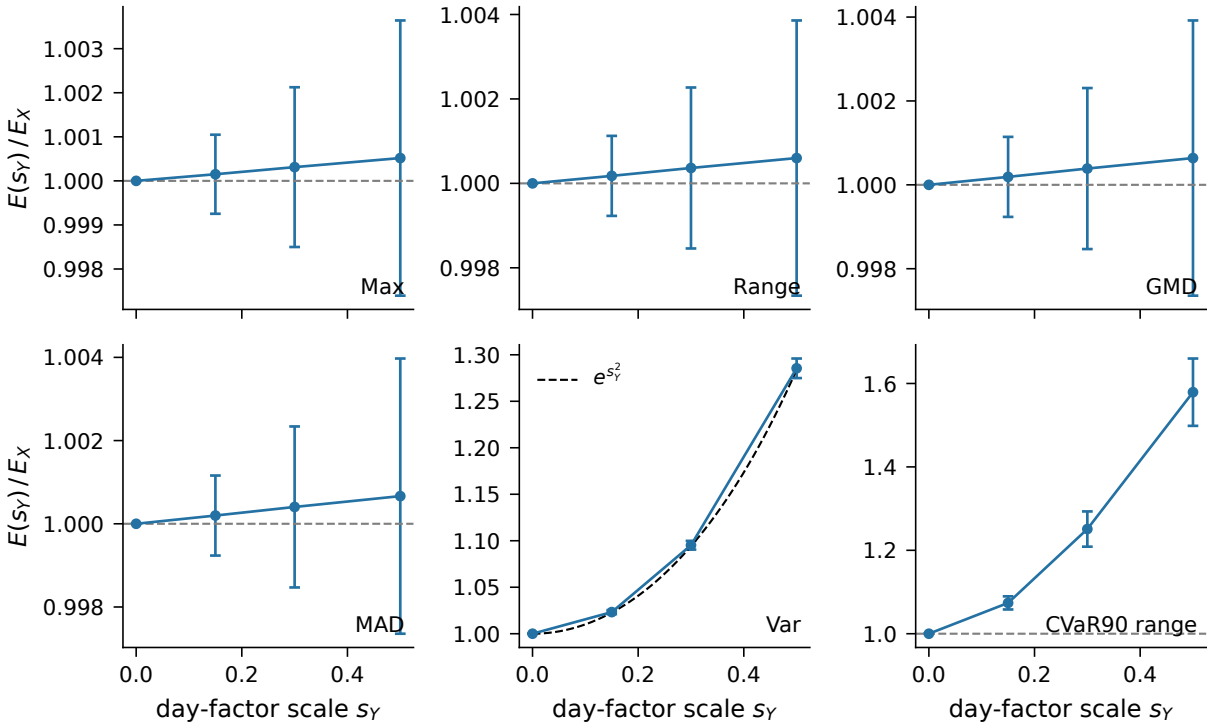


Figure S14: E11: effect of the day-factor dispersion s_Y on the expected ordered measures, the expected variance and the CVaR of the range.

Table S13: E12: noise-model misspecification. PBOI error of the gamma-based prediction against the true expected value under each family, and improvement of SORTIE-0 over Det evaluated under that family (format as Table S4).

Family	Source	Plans	PBOI |err| (%)	PBOI signed err. (%)	SORTIE-0 vs Det
Gamma	E3	500	0.16	+0.04	+1.23* (81/125)
Incident	E3	500	1.43	+1.35	+1.03* (71/125)
Lognormal	E3	500	0.29	+0.16	+1.21* (82/125)
Regional	E3	500	1.88	-0.74	+0.96* (65/125)
Gamma	E8	300	0.18	+0.10	+1.51* (53/75)
Lognormal	E8	300	0.51	+0.40	+1.48* (55/75)

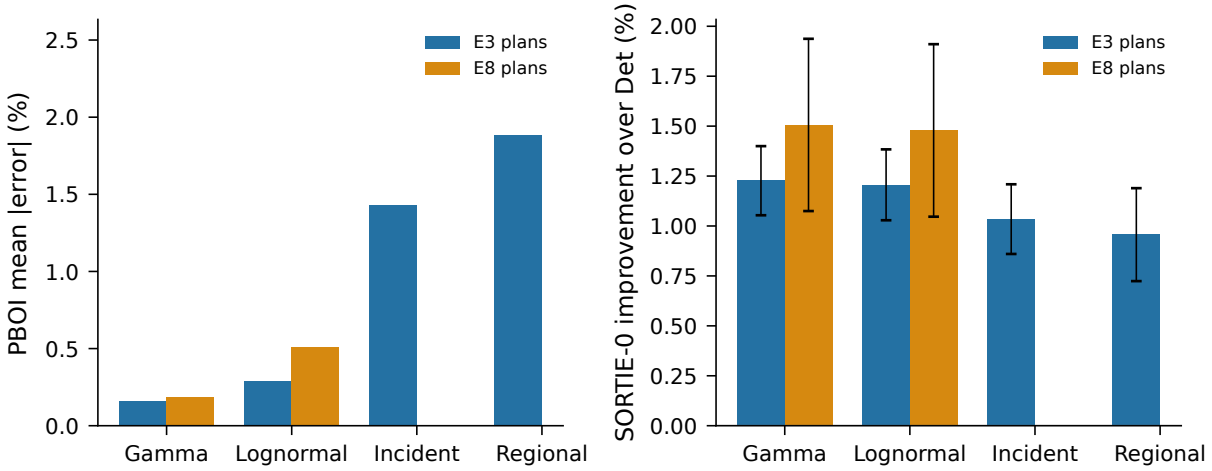


Figure S15: E12: PBOI error and improvement of SORTIE-0 over Det under four arc-noise families.

Table S14: E13: risk-consistent lower level, $n \in \{25, 35\}$, $B = 1.10$, ten instances per row. Routes changed: share of routes whose tour differs from the shortest tour of the same customers; extra length: total, per plan; mean risk: mean of $L_k + v\sqrt{W_k}$.

Measure	κ	Routes changed (%)	Extra length	Mean risk	Ex-post vs κ=0 [IQR]	Cost ratio
Range	0	0.0	0.0	1913.9	1.000 [1.000, 1.000]	1.098
Range	0.5	6.0	1.0	1912.9	1.000 [0.998, 1.000]	1.098
Range	1	4.0	0.9	1915.6	1.000 [0.995, 1.000]	1.098
Range	2	14.0	9.2	1914.0	1.000 [0.989, 1.004]	1.099
Range	4	14.0	14.1	1911.4	0.997 [0.989, 1.004]	1.099
Max	0	0.0	0.0	1890.7	1.000 [1.000, 1.000]	1.086
Max	0.5	2.0	0.5	1885.9	1.000 [1.000, 1.000]	1.084
Max	1	6.0	2.6	1891.3	1.000 [1.000, 1.000]	1.087
Max	2	6.0	3.9	1887.0	1.000 [1.000, 1.000]	1.085
Max	4	16.0	12.4	1882.6	1.000 [1.000, 1.000]	1.083

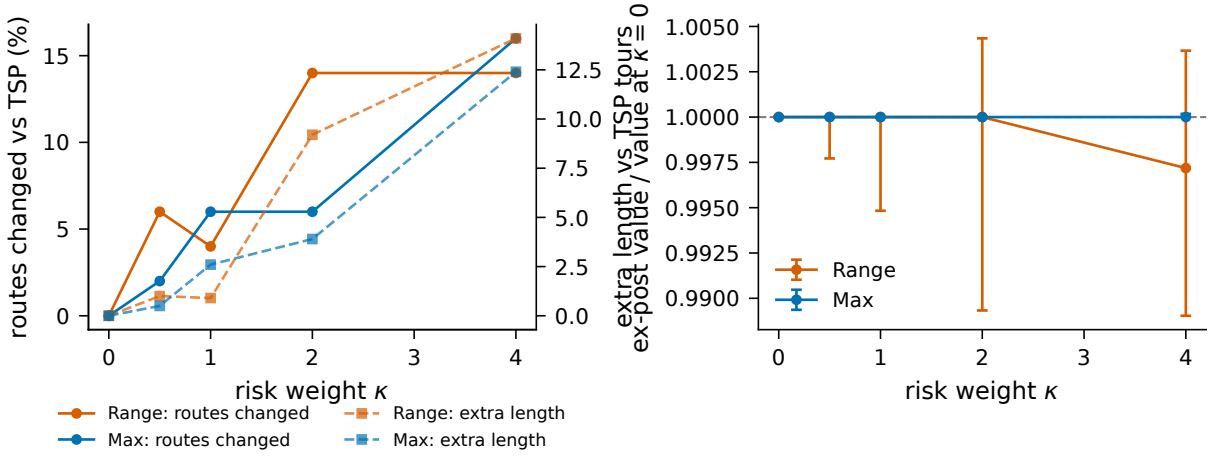


Figure S16: E13: effect of the drivers' risk weight κ.

Table S15: E14: rank-price prediction against exact evaluation. Groups: (instance, plan) pairs with at least one exactly improving move; top-8 hit: share of those groups in which the exact-best improving move is among the 8 best predicted moves (in brackets: expected share under a random order).

Measure	Plans	Moves	Groups with improving move	Spearman ρ	Top-8 hit (random)
Range	Cost plan	1448	10/10	0.93	0.80 (0.07)
Range	SORTIE-0 plan	346	1/10	0.80	0.00 (0.19)
Range	All plans	1794	11/20	0.92	0.73 (0.08)
GMD	Cost plan	1448	10/10	0.94	0.80 (0.07)
GMD	SORTIE-0 plan	333	0/10	0.78	– (–)
GMD	All plans	1781	10/20	0.94	0.80 (0.07)
Max	Cost plan	1448	10/10	0.89	0.80 (0.07)
Max	SORTIE-0 plan	554	1/10	0.89	0.00 (0.21)
Max	All plans	2002	11/20	0.89	0.73 (0.08)

Table S16: E14: number of candidates K evaluated per step; ex-post value relative to $K = 8$, median and interquartile range over 20 instance–measure pairs.

K	Range median [IQR]	GMD median [IQR]
1	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
2	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
4	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
8	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
16	1.000 [1.000, 1.000]	1.000 [0.996, 1.000]
32	1.000 [0.994, 1.000]	1.000 [1.000, 1.000]

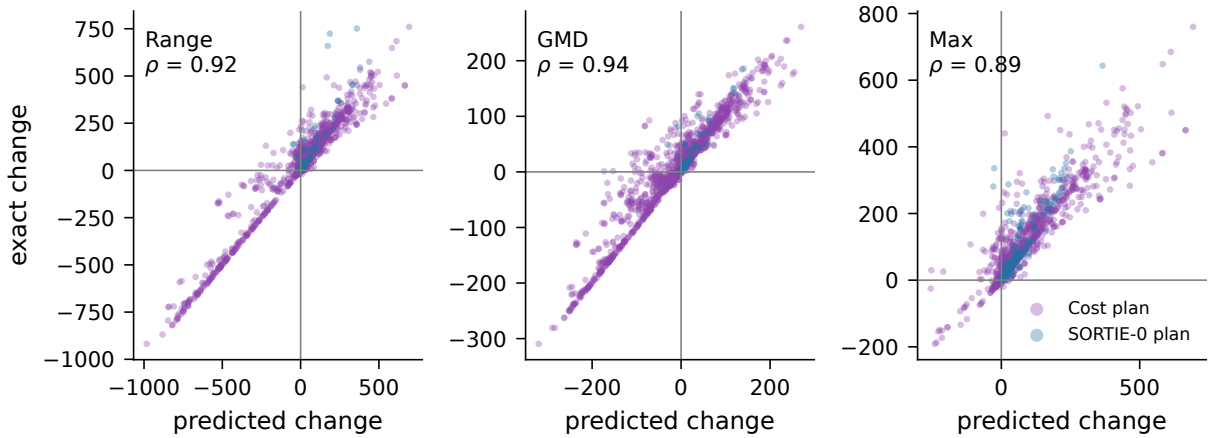


Figure S17: E14: predicted (rank prices) against exact change of the objective for 5577 moves.

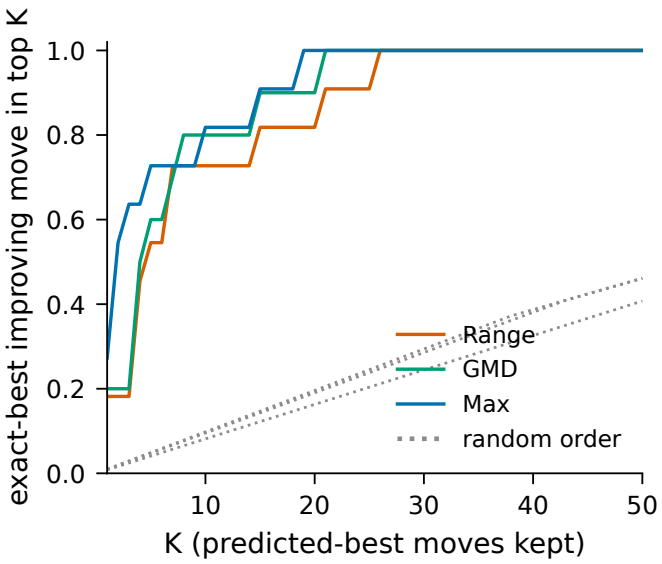


Figure S18: E14: share of cases in which the exact-best improving move is among the K best predicted moves.

Table S17: E15: improvement of SORTIE-0 over SAA with S scenarios, $n \in \{25, 35\}$, $B = 1.10$, ten instances per measure (format as Table S4; positive: SORTIE-0 better).

S	Range	GMD	Max	Pooled	p (pooled)
50	+0.17 (6/10)	+2.88* (8/10)	+0.99* (8/10)	+1.35* (22/30)	0.001
200	+2.11* (7/10)	+2.15* (7/10)	+0.46* (10/10)	+1.57* (24/30)	0.000
1000	+1.45 (5/10)	+1.64 (4/10)	+0.19 (6/10)	+1.09* (15/30)	0.002
5000	+1.34 (3/10)	+1.38 (2/10)	+0.02 (4/10)	+0.91 (9/30)	0.050

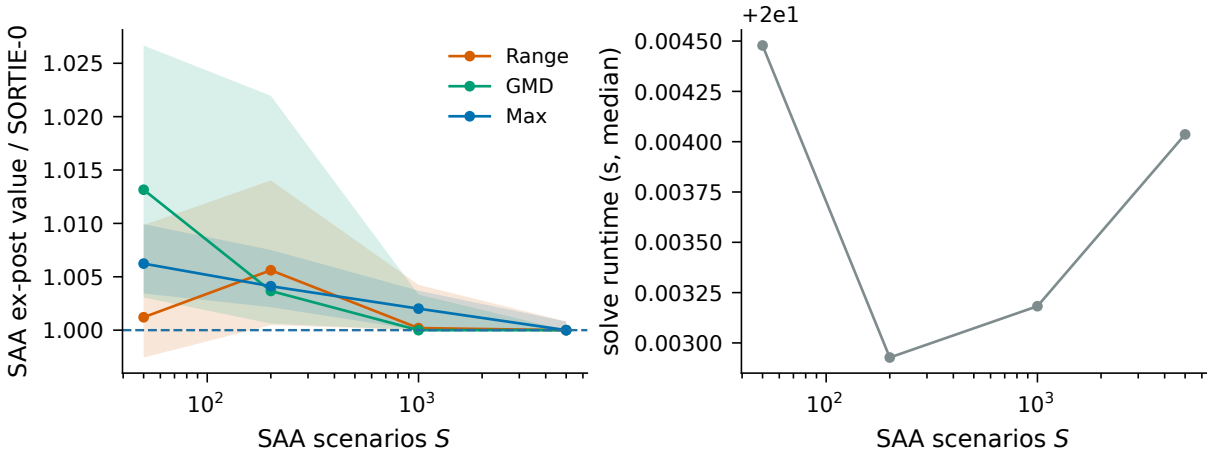


Figure S19: E15: ex-post value of SAA relative to SORTIE-0 against the sample size S.

Table S18: E16: risk-adjusted deterministic (Det-σ) and normal-marginal (PBOI-N) objectives, E3 grid, $B = 1.10$ (format as Table S4; the variance is omitted for PBOI-N because the expected empirical variance does not depend on the marginal family).

Measure	Det-σ vs Det	PBOI-N vs Det	SORTIE-0 vs Det	SORTIE-0 vs Det-σ
Range	-0.49 (8/25)	+0.48* (18/25)	+1.07* (17/25)	+1.51* (19/25)
GMD	-0.69 (8/25)	+0.83* (14/25)	+0.94* (14/25)	+1.50* (18/25)
Max	-0.08 (8/25)	+2.08* (24/25)	+2.08* (24/25)	+2.13* (25/25)
Pooled	-0.81† (37/125)	+1.10* (71/100)	+1.23* (81/125)	+1.86* (97/125)

Table S19: E17: second benchmark family (12 X instances, depot plus first 50 customers), $B = 1.10$, 120 s, mean of two solver seeds; improvement of the ex-post value over Det (format as Table S4).

Measure	SAA vs Det	SORTIE-0 vs Det	Det-σ vs Det	Det belief err. (%)
Range	+1.48 (8/12)	+2.91* (11/12)	-0.39 (5/12)	-40.3
GMD	-0.62 (5/12)	+1.64 (8/12)	-2.15† (1/12)	-34.7
Max	+1.36* (10/12)	+1.93* (12/12)	+0.01 (7/12)	-8.4
Pooled	+0.74* (23/36)	+2.16* (31/36)	-0.84 (13/36)	-27.8

Table S20: E17 per instance: depot position, C^* and the improvement of SORTIE-0 over Det pooled over the three measures.

Instance	Depot	C^*	SORTIE-0 vs Det, mean of measures (wins/3)
X-n101-k25-h50	intermediate	7415	+2.80 (3/3)
X-n106-k14-h50	eccentric	9767	+2.02 (3/3)
X-n110-k13-h50	central	7194	+3.65 (3/3)
X-n115-k10-h50	central	7183	+2.04 (3/3)
X-n120-k6-h50	eccentric	9809	+1.89 (3/3)
X-n125-k30-h50	eccentric	9842	+1.58 (3/3)
X-n129-k18-h50	eccentric	9821	+1.03 (2/3)
X-n134-k13-h50	eccentric	4686	+0.69 (2/3)
X-n139-k10-h50	central	7253	+4.24 (3/3)
X-n143-k7-h50	eccentric	10398	+2.94 (2/3)
X-n148-k46-h50	intermediate	7752	+1.76 (2/3)
X-n153-k22-h50	intermediate	5786	+1.27 (2/3)

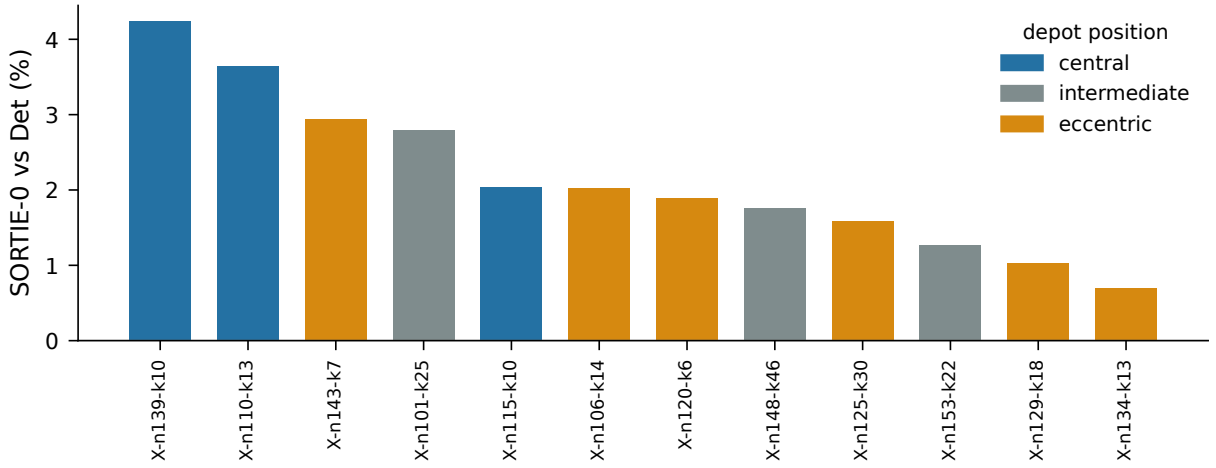


Figure S20: E17: improvement of SORTIE-0 over Det per instance (mean over measures and seeds), coloured by depot position.

S3 Per-instance results

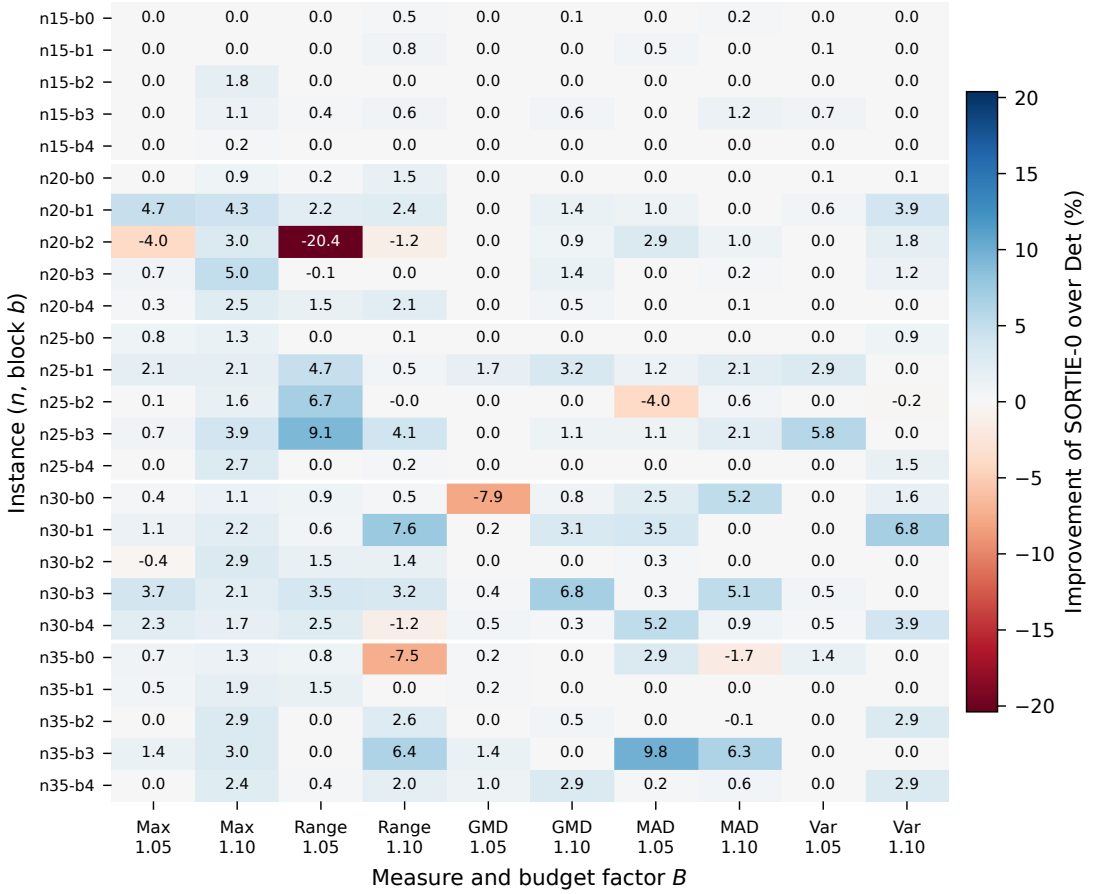


Figure S21: E3: improvement of the ex-post value of SORTIE-0 over Det (%) for every instance, measure and budget.

Tables S21 and S22 list the out-of-sample ex-post value of every E3 plan, Table S23 the seed means and standard deviations of E7b, and Table S24 the E17 values per instance and measure . Every plan is stored in the released plan cache and can be re-evaluated with the released simulator.

Table S21: E3, $B = 1.10$: out-of-sample ex-post value of every plan (bold: best); Cost: the cost-optimal plan; S-0/S-1: SORTIE-0/1; Gain: improvement of S-0 over Det; last column: total length of the S-0 plan relative to C^*.

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n15-b0	Max	2,640	2,547	2,561	2,547	**2,545**	+0.00	1.026
	Range	2,412	1,575	**1,568**	**1,568**	1,573	+0.47	1.098
	GMD	907	625	625	625	**623**	+0.06	1.100
	MAD	652	468	**467**	**467**	467	+0.20	1.100
	Var	730,909	357,317	357,317	357,238	**354,022**	+0.02	1.100
n15-b1	Max	2,165	2,091	2,095	**2,091**	**2,091**	+0.00	1.098
	Range	1,875	1,295	**1,285**	**1,285**	**1,285**	+0.79	1.095
	GMD	729	**530**	**530**	**530**	**530**	+0.00	1.095
	MAD	537	**405**	**405**	**405**	**405**	+0.00	1.095
	Var	461,317	**258,741**	261,374	**258,741**	**258,741**	+0.00	1.095
n15-b2	Max	2,419	2,419	2,376	**2,376**	**2,376**	+1.78	1.070

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	Range	2,187	**1,231**	1,234	**1,231**	**1,231**	+0.00	1.081
	GMD	804	**478**	482	**478**	479	+0.00	1.099
	MAD	568	**347**	**347**	**347**	348	+0.00	1.099
	Var	604,666	**222,814**	223,451	**222,814**	223,451	+0.00	1.099
n15-b3	Max	1,926	1,826	1,826	1,805	**1,803**	+1.13	1.070
	Range	1,899	784	806	779	**773**	+0.64	1.070
	GMD	665	305	305	303	**300**	+0.59	1.070
	MAD	452	226	224	223	**221**	+1.15	1.070
	Var	471,206	90,921	95,342	90,921	**89,222**	+0.00	1.095
n15-b4	Max	2,412	1,940	**1,934**	1,935	1,935	+0.25	1.056
	Range	2,170	**1,024**	**1,024**	**1,024**	**1,024**	+0.00	1.098
	GMD	772	**417**	**417**	**417**	**417**	+0.00	1.098
	MAD	531	**316**	**316**	**316**	**316**	+0.00	1.098
	Var	543,628	**164,264**	**164,264**	**164,264**	**164,264**	+0.00	1.098
n20-b0	Max	2,656	2,651	2,647	**2,628**	**2,628**	+0.90	1.081
	Range	2,428	1,549	1,537	1,526	**1,522**	+1.48	1.097
	GMD	926	628	631	628	**626**	+0.00	1.099
	MAD	672	477	481	477	**477**	+0.00	1.099
	Var	782,670	358,873	361,706	358,394	**356,722**	+0.13	1.097
n20-b1	Max	1,756	1,743	**1,668**	1,669	1,669	+4.28	1.037
	Range	1,332	563	562	550	**550**	+2.42	1.099
	GMD	472	217	217	214	**214**	+1.40	1.077
	MAD	323	157	159	**157**	**157**	+0.04	1.077
	Var	230,329	47,975	47,125	**46,117**	**46,117**	+3.87	1.077
n20-b2	Max	2,393	2,224	2,177	**2,158**	**2,158**	+2.97	1.096
	Range	2,309	**1,283**	**1,283**	1,299	1,293	-1.24	1.090
	GMD	886	558	**552**	553	553	+0.92	1.099
	MAD	645	387	**383**	383	383	+1.02	1.099
	Var	692,985	257,489	**252,913**	**252,913**	**252,913**	+1.78	1.098
n20-b3	Max	2,668	2,472	2,349	**2,348**	**2,348**	+5.02	1.084
	Range	2,518	**1,272**	1,278	**1,272**	**1,272**	+0.00	1.100
	GMD	918	523	517	**516**	523	+1.42	1.099
	MAD	644	373	**372**	**372**	374	+0.17	1.099
	Var	757,430	250,084	247,116	247,116	**245,946**	+1.19	1.098
n20-b4	Max	1,981	1,823	1,782	1,778	**1,776**	+2.48	1.068
	Range	1,472	1,085	1,066	1,062	**1,060**	+2.13	1.068
	GMD	617	399	398	397	**396**	+0.52	1.067
	MAD	477	282	**282**	**282**	**282**	+0.14	1.098
	Var	332,502	147,358	148,357	147,358	**146,846**	+0.00	1.067
n25-b0	Max	2,607	2,622	2,613	**2,588**	**2,588**	+1.30	1.095
	Range	2,379	1,253	1,260	**1,252**	1,253	+0.11	1.100
	GMD	858	**506**	508	**506**	507	+0.00	1.100
	MAD	597	381	383	381	**381**	+0.00	1.100
	Var	718,676	245,392	248,480	**243,295**	**243,295**	+0.85	1.099
n25-b1	Max	2,145	1,971	1,939	**1,931**	**1,931**	+2.05	1.027
	Range	1,257	618	615	615	**611**	+0.54	1.098
	GMD	535	247	238	239	**236**	+3.19	1.098
	MAD	417	179	176	175	**173**	+2.08	1.098
	Var	255,701	58,275	60,272	58,275	**57,003**	+0.00	1.098
n25-b2	Max	2,148	2,065	2,035	**2,033**	**2,033**	+1.56	1.097
	Range	2,036	**1,312**	1,312	1,312	1,312	-0.01	1.099
	GMD	763	539	539	539	**537**	+0.00	1.092
	MAD	546	412	409	409	**408**	+0.62	1.092
	Var	516,255	265,863	267,817	266,385	**264,172**	-0.20	1.099
n25-b3	Max	1,984	2,025	1,958	**1,946**	**1,946**	+3.90	1.097
	Range	1,474	874	840	838	**836**	+4.13	1.099

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	GMD	560	331	333	**328**	331	+1.06	1.098
	MAD	405	248	244	**242**	**242**	+2.15	1.098
	Var	283,090	**104,507**	107,279	**104,507**	105,353	+0.00	1.098
n25-b4	Max	2,151	2,197	2,143	2,137	**2,137**	+2.74	1.099
	Range	1,931	1,002	1,004	999	**994**	+0.21	1.099
	GMD	683	393	394	393	**391**	+0.00	1.100
	MAD	468	291	292	291	**290**	+0.00	1.100
	Var	459,013	152,985	153,860	150,758	**149,397**	+1.46	1.099
n30-b0	Max	2,773	2,615	2,587	**2,585**	**2,585**	+1.14	1.050
	Range	2,626	1,229	1,228	1,223	**1,223**	+0.47	1.099
	GMD	1,000	497	497	**493**	493	+0.79	1.099
	MAD	724	392	372	**371**	372	+5.22	1.099
	Var	911,746	235,555	233,126	**231,861**	**231,861**	+1.57	1.099
n30-b1	Max	2,657	2,574	2,540	**2,516**	2,521	+2.23	1.075
	Range	2,287	1,537	1,421	**1,420**	**1,420**	+7.63	1.095
	GMD	913	568	551	551	**550**	+3.06	1.100
	MAD	684	411	**409**	411	412	+0.00	1.098
	Var	730,545	297,591	298,366	277,261	**275,214**	+6.83	1.095
n30-b2	Max	1,957	1,838	1,784	1,784	**1,783**	+2.94	1.078
	Range	1,838	1,002	988	988	**961**	+1.44	1.091
	GMD	667	365	372	365	**365**	+0.00	1.091
	MAD	466	259	263	259	**259**	+0.00	1.091
	Var	419,940	126,369	126,369	126,369	**126,081**	+0.00	1.091
n30-b3	Max	2,401	2,350	2,307	**2,300**	**2,300**	+2.13	1.097
	Range	2,371	1,264	**1,223**	**1,223**	1,284	+3.25	1.100
	GMD	853	532	502	**496**	500	+6.75	1.100
	MAD	592	396	**369**	375	**369**	+5.12	1.100
	Var	717,888	**225,491**	229,750	**225,491**	228,530	+0.00	1.100
n30-b4	Max	2,325	2,231	2,196	2,193	**2,191**	+1.69	1.068
	Range	2,100	**1,217**	1,237	1,232	1,231	-1.21	1.093
	GMD	762	494	502	493	**492**	+0.27	1.098
	MAD	533	347	351	344	**344**	+0.90	1.100
	Var	536,206	225,051	219,695	**216,280**	225,293	+3.90	1.099
n35-b0	Max	2,892	2,717	2,704	**2,682**	**2,682**	+1.27	1.098
	Range	2,744	**1,376**	1,741	1,479	1,724	-7.46	1.096
	GMD	1,013	**529**	609	**529**	609	+0.00	1.100
	MAD	718	421	429	428	**385**	-1.72	1.098
	Var	964,248	**259,530**	372,532	**259,530**	368,171	+0.00	1.100
n35-b1	Max	2,543	2,127	2,090	**2,087**	**2,087**	+1.91	1.096
	Range	2,429	**825**	**825**	**825**	**825**	+0.00	1.098
	GMD	946	**323**	324	**323**	**323**	+0.00	1.098
	MAD	696	**238**	239	**238**	**238**	+0.00	1.098
	Var	758,306	**102,506**	102,787	**102,506**	**102,506**	+0.00	1.098
n35-b2	Max	1,922	1,794	1,742	**1,741**	1,814	+2.91	1.061
	Range	1,526	553	546	**538**	**538**	+2.61	1.093
	GMD	562	209	**208**	**208**	**208**	+0.48	1.093
	MAD	398	**153**	153	153	153	-0.06	1.093
	Var	296,372	43,908	43,424	**42,644**	**42,644**	+2.88	1.095
n35-b3	Max	2,527	2,452	2,398	2,379	**2,379**	+2.98	1.100
	Range	2,137	1,066	1,012	**998**	**998**	+6.37	1.100
	GMD	780	**377**	381	**377**	**377**	+0.00	1.100
	MAD	548	291	273	273	**273**	+6.35	1.100
	Var	555,805	**142,628**	144,945	**142,628**	**142,628**	+0.00	1.100
n35-b4	Max	2,510	2,186	2,142	**2,133**	**2,133**	+2.43	1.085
	Range	2,436	702	702	**687**	**687**	+2.03	1.097
	GMD	902	272	275	**264**	**264**	+2.92	1.097

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	MAD	640	194	**193**	**193**	**193**	+0.58	1.097
	Var	772,452	71,590	72,810	**69,523**	**69,523**	+2.89	1.097

Table S22: E3, $B = 1.05$: out-of-sample ex-post value of every plan (bold: best); Cost: the cost-optimal plan; S-0/S-1: SORTIE-0/1; Gain: improvement of S-0 over Det; last column: total length of the S-0 plan relative to C^*.

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n15-b0	Max	2,640	**2,547**	2,559	**2,547**	**2,547**	+0.00	1.026
	Range	2,412	**2,319**	2,331	**2,319**	**2,319**	+0.00	1.026
	GMD	907	**868**	873	**868**	**868**	+0.00	1.026
	MAD	652	**621**	**621**	**621**	**621**	+0.00	1.026
	Var	730,909	**691,734**	699,905	**691,734**	**691,734**	+0.00	1.026
n15-b1	Max	2,165	**2,091**	**2,091**	**2,091**	**2,091**	+0.00	1.041
	Range	1,875	**1,503**	**1,503**	**1,503**	1,505	+0.00	1.040
	GMD	729	**616**	**616**	**616**	617	+0.00	1.040
	MAD	537	469	469	**467**	468	+0.50	1.048
	Var	461,317	337,238	339,053	**336,907**	337,815	+0.10	1.048
n15-b2	Max	2,419	**2,419**	**2,419**	**2,419**	**2,419**	+0.00	1.000
	Range	2,187	**2,187**	**2,187**	**2,187**	**2,187**	+0.00	1.000
	GMD	804	**804**	**804**	**804**	**804**	+0.00	1.000
	MAD	568	**568**	**568**	**568**	**568**	+0.00	1.000
	Var	604,666	**604,666**	**604,666**	**604,666**	**604,666**	+0.00	1.000
n15-b3	Max	1,926	**1,926**	**1,926**	**1,926**	**1,926**	+0.00	1.000
	Range	1,899	1,181	**1,177**	**1,177**	**1,177**	+0.37	1.043
	GMD	665	468	**468**	468	468	+0.00	1.041
	MAD	452	**349**	349	**349**	**349**	+0.00	1.041
	Var	471,206	200,220	**198,775**	**198,775**	**198,775**	+0.72	1.043
n15-b4	Max	2,412	**1,940**	**1,940**	**1,940**	**1,940**	+0.00	1.000
	Range	2,170	**1,698**	**1,698**	**1,698**	**1,698**	+0.00	1.000
	GMD	772	**648**	**648**	**648**	**648**	+0.00	1.022
	MAD	531	**466**	**466**	**466**	**466**	+0.00	1.022
	Var	543,628	**375,246**	**375,246**	**375,246**	**375,246**	+0.00	1.000
n20-b0	Max	2,656	**2,639**	2,647	**2,639**	**2,639**	+0.00	1.021
	Range	2,428	2,411	2,418	**2,405**	**2,405**	+0.25	1.022
	GMD	926	**863**	**863**	**863**	**863**	+0.00	1.049
	MAD	672	**595**	**595**	**595**	**595**	+0.00	1.049
	Var	782,670	748,918	751,999	**748,239**	**748,239**	+0.09	1.022
n20-b1	Max	1,756	1,751	**1,668**	1,669	1,669	+4.72	1.037
	Range	1,332	666	**652**	**652**	**652**	+2.20	1.036
	GMD	472	251	**251**	251	251	+0.00	1.049
	MAD	323	185	184	**183**	**183**	+0.99	1.049
	Var	230,329	62,141	**61,759**	**61,759**	**61,759**	+0.62	1.036
n20-b2	Max	2,393	**2,191**	2,202	2,279	2,279	-4.03	1.039
	Range	2,309	**1,652**	1,988	1,988	1,988	-20.38	1.039
	GMD	886	**656**	**656**	**656**	**656**	+0.00	1.049
	MAD	645	505	**490**	**490**	**490**	+2.92	1.049
	Var	692,985	**379,751**	**379,751**	**379,751**	**379,751**	+0.00	1.049
n20-b3	Max	2,668	2,398	**2,381**	**2,381**	**2,381**	+0.71	1.046
	Range	2,518	**1,640**	1,642	1,642	1,642	-0.10	1.044
	GMD	918	**673**	**673**	**673**	**673**	+0.00	1.046
	MAD	644	**490**	**490**	**490**	**490**	+0.00	1.046
	Var	757,430	**408,393**	**408,393**	**408,393**	**408,393**	+0.00	1.046

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
n20-b4	Max	1,981	1,819	**1,814**	**1,814**	1,815	+0.30	1.050
	Range	1,472	1,235	**1,216**	**1,216**	**1,216**	+1.51	1.047
	GMD	617	**485**	**485**	**485**	**485**	+0.00	1.048
	MAD	477	**341**	**341**	**341**	**341**	+0.00	1.048
	Var	332,502	208,876	208,876	208,876	**208,721**	+0.00	1.045
n25-b0	Max	2,607	2,610	2,613	**2,589**	**2,589**	+0.81	1.019
	Range	2,379	**1,755**	**1,755**	**1,755**	**1,755**	+0.00	1.050
	GMD	858	**683**	**683**	**683**	**683**	+0.00	1.050
	MAD	597	**503**	**503**	**503**	**503**	+0.00	1.050
	Var	718,676	**416,021**	**416,021**	**416,021**	**416,021**	+0.00	1.050
n25-b1	Max	2,145	1,971	1,938	**1,931**	**1,931**	+2.05	1.027
	Range	1,257	774	745	**738**	**738**	+4.66	1.027
	GMD	535	290	288	**285**	**285**	+1.69	1.027
	MAD	417	212	211	**209**	**209**	+1.22	1.027
	Var	255,701	82,851	81,533	**80,430**	**80,430**	+2.92	1.027
n25-b2	Max	2,148	2,148	2,147	**2,145**	**2,145**	+0.13	1.029
	Range	2,036	1,815	**1,693**	**1,693**	**1,693**	+6.71	1.046
	GMD	763	**654**	**654**	**654**	689	+0.00	1.046
	MAD	546	**479**	504	498	**479**	-3.99	1.040
	Var	516,255	418,916	**372,661**	418,916	417,741	+0.00	1.036
n25-b3	Max	1,984	1,968	1,958	**1,953**	**1,953**	+0.72	1.045
	Range	1,474	1,162	1,058	**1,056**	**1,056**	+9.13	1.045
	GMD	560	**419**	423	**419**	**419**	+0.00	1.046
	MAD	405	314	312	**311**	**311**	+1.10	1.050
	Var	283,090	169,897	**160,109**	**160,109**	**160,109**	+5.76	1.046
n25-b4	Max	2,151	2,151	2,151	**2,150**	**2,150**	+0.03	1.044
	Range	1,931	**1,433**	1,544	**1,433**	1,544	+0.00	1.034
	GMD	683	**565**	**565**	**565**	**565**	+0.00	1.044
	MAD	468	**397**	**397**	**397**	**397**	+0.00	1.044
	Var	459,013	**294,618**	**294,618**	**294,618**	**294,618**	+0.00	1.044
n30-b0	Max	2,773	2,606	2,602	**2,597**	2,597	+0.36	1.044
	Range	2,626	2,388	2,393	2,368	**2,357**	+0.86	1.044
	GMD	1,000	**760**	831	820	821	-7.91	1.044
	MAD	724	566	551	552	**548**	+2.54	1.044
	Var	911,746	516,782	516,782	516,782	**515,564**	+0.00	1.048
n30-b1	Max	2,657	2,552	2,530	**2,524**	2,526	+1.13	1.037
	Range	2,287	1,903	1,893	**1,892**	1,893	+0.61	1.041
	GMD	913	759	761	758	**706**	+0.16	1.044
	MAD	684	535	516	**516**	517	+3.54	1.049
	Var	730,545	**460,952**	**460,952**	**460,952**	**460,952**	+0.00	1.044
n30-b2	Max	1,957	**1,942**	1,958	1,950	1,949	-0.39	1.034
	Range	1,838	1,224	1,217	1,205	**1,204**	+1.53	1.049
	GMD	667	508	514	508	**508**	+0.00	1.048
	MAD	466	393	393	**392**	394	+0.28	1.049
	Var	419,940	225,689	230,414	225,689	**225,408**	+0.00	1.048
n30-b3	Max	2,401	2,387	2,306	**2,300**	**2,300**	+3.67	1.042
	Range	2,371	1,946	**1,878**	**1,878**	**1,878**	+3.51	1.050
	GMD	853	664	**661**	**661**	**661**	+0.39	1.050
	MAD	592	453	453	**451**	**451**	+0.32	1.050
	Var	717,888	449,606	**447,388**	**447,388**	**447,388**	+0.49	1.050
n30-b4	Max	2,325	2,244	2,200	**2,193**	**2,193**	+2.28	1.045
	Range	2,100	1,739	1,701	**1,695**	**1,695**	+2.53	1.043
	GMD	762	632	632	**629**	630	+0.51	1.050
	MAD	533	457	**433**	**433**	**433**	+5.16	1.048
	Var	536,206	354,473	364,184	**352,875**	**352,875**	+0.45	1.046
n35-b0	Max	2,892	2,717	2,703	2,697	**2,694**	+0.75	1.035

Instance	Measure	Cost	Det	SAA	S-0	S-1	Gain (%)	$\sum L/C^*$
	Range	2,744	2,488	2,486	2,468	**2,447**	+0.80	1.035
	GMD	1,013	846	860	844	**841**	+0.24	1.028
	MAD	718	580	574	564	**562**	+2.88	1.028
	Var	964,248	809,215	800,421	798,285	**793,815**	+1.35	1.022
n35-b1	Max	2,543	2,245	**2,137**	2,233	**2,137**	+0.52	1.045
	Range	2,429	1,793	1,768	1,767	**1,239**	+1.48	1.049
	GMD	946	502	502	501	**499**	+0.22	1.050
	MAD	696	**374**	**374**	**374**	**374**	+0.00	1.050
	Var	758,306	228,342	230,146	228,342	**228,317**	+0.00	1.050
n35-b2	Max	1,922	**1,748**	**1,748**	**1,748**	**1,748**	+0.00	1.047
	Range	1,526	**620**	**620**	**620**	**620**	+0.00	1.047
	GMD	562	**246**	**246**	**246**	**246**	+0.00	1.047
	MAD	398	**184**	**184**	**184**	**184**	+0.00	1.047
	Var	296,372	**58,746**	**58,746**	**58,746**	**58,746**	+0.00	1.047
n35-b3	Max	2,527	2,453	2,420	**2,420**	2,423	+1.36	1.037
	Range	2,137	**1,292**	**1,292**	**1,292**	**1,292**	+0.00	1.050
	GMD	780	532	**524**	**524**	**524**	+1.42	1.050
	MAD	548	440	**397**	**397**	**397**	+9.77	1.050
	Var	555,805	**247,798**	**247,798**	**247,798**	**247,798**	+0.00	1.050
n35-b4	Max	2,510	**2,153**	2,174	**2,153**	**2,153**	+0.00	1.025
	Range	2,436	1,026	**1,022**	**1,022**	**1,022**	+0.38	1.050
	GMD	902	415	**411**	**411**	**411**	+0.96	1.050
	MAD	640	290	**290**	**290**	**290**	+0.24	1.048
	Var	772,452	**156,368**	158,633	**156,368**	158,899	+0.00	1.050

Table S23: E7b, $n = 100$: mean (standard deviation) over three solver seeds of the ex-post value.

Instance	Measure	Det	SAA	S-0	S-0 vs Det (%)
n100-b0	GMD	378 (53.9)	349 (11.7)	337 (3.7)	+10.77
n100-b0	Max	3,016 (137)	2,930 (6.2)	2,924 (6.3)	+3.06
n100-b0	Range	870 (1.0)	861 (12.9)	840 (0.3)	+3.44
n100-b1	GMD	249 (30.4)	231 (1.6)	235 (4.6)	+5.71
n100-b1	Max	2,727 (9.7)	2,665 (2.7)	2,667 (0.3)	+2.20
n100-b1	Range	604 (3.5)	648 (27.8)	610 (8.6)	-1.05
n100-b2	GMD	346 (17.6)	351 (6.7)	401 (39.8)	-15.87
n100-b2	Max	2,444 (8.5)	2,493 (32.0)	2,484 (62.5)	-1.65
n100-b2	Range	1,081 (318)	929 (44.4)	970 (5.4)	+10.26
n100-b3	GMD	304 (10.3)	313 (14.9)	305 (7.8)	-0.20
n100-b3	Max	2,583 (51.7)	2,535 (0.2)	2,602 (117)	-0.72
n100-b3	Range	881 (10.6)	785 (7.3)	799 (9.8)	+9.35
n100-b4	GMD	516 (5.5)	523 (19.8)	502 (11.3)	+2.79
n100-b4	Max	2,525 (8.6)	2,497 (12.4)	2,475 (15.8)	+2.02
n100-b4	Range	1,346 (64.1)	1,365 (63.8)	1,335 (50.6)	+0.83
n100-b5	GMD	289 (6.6)	273 (4.1)	243 (22.8)	+15.89
n100-b5	Max	2,396 (12.8)	2,431 (62.3)	2,391 (8.5)	+0.23
n100-b5	Range	890 (282)	700 (85.3)	630 (69.9)	+29.24

Table S24: E17: ex-post value, mean over two solver seeds (bold: best).

Instance	Measure	Det	SAA	S-0	Det-σ	S-0 vs Det (%)
X-n101-k25-h50	GMD	267	268	**265**	279	+0.65
	Max	2,049	2,027	**2,013**	2,060	+1.73
	Range	755	731	**709**	767	+6.01
X-n106-k14-h50	GMD	499	508	**497**	513	+0.36
	Max	2,874	2,821	**2,811**	2,869	+2.19
	Range	1,317	1,285	**1,271**	1,300	+3.52
X-n110-k13-h50	GMD	154	152	**148**	156	+4.21
	Max	1,781	1,731	**1,728**	1,751	+2.97
	Range	397	392	**382**	395	+3.76
X-n115-k10-h50	GMD	167	160	**159**	174	+4.67
	Max	1,788	1,790	**1,781**	1,800	+0.40
	Range	409	415	**405**	414	+1.07
X-n120-k6-h50	GMD	609	662	**583**	636	+4.32
	Max	2,843	2,830	**2,829**	2,859	+0.51
	Range	1,493	1,499	**1,480**	1,518	+0.83
X-n125-k30-h50	GMD	358	359	**352**	365	+1.69
	Max	2,689	2,670	**2,642**	2,696	+1.75
	Range	929	929	916	**914**	+1.30
X-n129-k18-h50	GMD	**557**	581	565	560	-1.31
	Max	2,843	2,817	**2,795**	2,842	+1.69
	Range	1,437	1,399	**1,398**	1,417	+2.72
X-n134-k13-h50	GMD	460	**460**	460	463	+0.05
	Max	1,935	1,936	**1,894**	1,986	+2.08
	Range	**1,354**	1,358	1,355	1,378	-0.06
X-n139-k10-h50	GMD	173	160	**159**	164	+8.01
	Max	1,804	1,788	**1,785**	1,792	+1.07
	Range	428	413	**412**	432	+3.63
X-n143-k7-h50	GMD	536	**535**	544	550	-1.52
	Max	3,087	2,979	**2,960**	3,038	+4.10
	Range	1,442	1,365	**1,352**	1,482	+6.24
X-n148-k46-h50	GMD	**250**	252	253	258	-0.95
	Max	2,087	2,021	**2,017**	2,080	+3.33
	Range	600	597	**583**	586	+2.88
X-n153-k22-h50	GMD	**177**	186	178	188	-0.54
	Max	1,523	1,506	**1,504**	1,517	+1.28
	Range	472	466	**457**	480	+3.06

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